

Module 1: Comprehensive Formula Sheet

Finance – Key Mathematical Models and Numerical Examples

Contents

1	Time Value of Money (TVM) & Cash Flow Valuation	7
1.1	Future Value – One Period	7
1.2	Future Value – Multiple Periods	7
1.3	Present Value – One Period	7
1.4	Present Value – Multiple Periods	8
1.5	Time to Reach a Future Value	8
1.6	Present Value of an Annuity	8
1.7	Present Value of a Perpetuity	8
1.8	Present Value of a Growing Perpetuity	9
2	Compounding & Interest Rates	9
2.1	Fisher Equation	9
2.2	APR vs. EAR	10
2.3	Continuous Compounding	10
3	Risk, Return & Basic Statistics	11
3.1	Expected Return	11
3.2	Variance and Standard Deviation	11
3.3	Historical Average Return	12
3.4	Historical Variance – Sample Variance	12
3.5	Covariance and Correlation	12
4	Portfolio Theory & Diversification	13
4.1	Portfolio Return	13
4.2	Portfolio Variance – Two Assets	13
4.3	Variance of an Equally Weighted Portfolio	14
5	Asset Pricing, CAPM & Risk Metrics	15
5.1	Beta	15
5.2	CAPM – Expected Return	15
5.3	Alpha	15
5.4	Portfolio Beta	16
5.5	Sharpe Ratio	16
6	Market Microstructure & Execution Cost	16
6.1	Bid-Ask Spread and Mid-Price	16
6.2	Percentage Spread	17
6.3	Volume-Weighted Average Price (VWAP)	17
7	Corporate Valuation & Free Cash Flow	18
7.1	Free Cash Flow to the Firm	18

8	Summary Tips for Exams	18
8.1	Discounting Treatment	18
8.2	Cash Flow Timelines	18
8.3	Risk Diversification	19
9	Fintech Economics, Credit Risk & Model Evaluation	20
9.1	Customer Acquisition Cost (CAC) and Lifetime Value (LTV)	20
9.1.1	Concept	20
9.1.2	Key Formulas	20
9.1.3	Worked Example	21
9.2	Logistic Regression – The Baseline Underwriting Model	22
9.2.1	Concept	22
9.2.2	Logit Form	22
9.2.3	Probability Form	23
9.2.4	Meaning of the Variables	23
9.2.5	Worked Example	23
9.3	Confusion Matrix Evaluation: Sensitivity and Specificity	24
9.3.1	Concept	24
9.3.2	Four Possible Outcomes	25
9.3.3	Sensitivity	25
9.3.4	Specificity	25
9.3.5	Worked Example	25
9.4	Kolmogorov–Smirnov (KS) Statistic	26
9.4.1	Concept	26
9.4.2	KS Distance	26
9.4.3	Worked Example	27
9.5	Expected Loss Minimization and Score Cut-Offs	28
9.5.1	Concept	28
9.5.2	Net Profit Formula	29
9.5.3	Worked Example	29
9.6	Peer-to-Peer (P2P) Lending Platform Calculations	30
9.6.1	Concept	30
9.6.2	Interest Earned	30
9.6.3	Platform Spread	30
9.6.4	Lender Net Return	31
9.6.5	Worked Example	31
9.7	Quick Formula Reference	32
9.8	Key Takeaways	32
10	Financial Analytics: Formula Sheet and Strategic Reference Guide	33
10.1	Descriptive Statistics: Measuring Central Tendency and Dispersion	33
10.1.1	Arithmetic Mean	33
10.1.2	Weighted Mean	33
10.1.3	Median	34
10.1.4	Percentiles and Quartiles	34
10.1.5	Range	34
10.1.6	Interquartile Range	34
10.1.7	Variance	34

10.1.8	Standard Deviation	35
10.1.9	Coefficient of Variation	35
10.1.10	Strategic Interpretation: Risk versus Stability	35
10.1.11	Step-by-Step Example: Weighted Portfolio Return	35
10.2	Distributions and Shape: The Geometry of Risk	36
10.2.1	Z-Score	36
10.2.2	Empirical Rule	36
10.2.3	Skewness	36
10.2.4	Strategic Interpretation: Downside Risk	37
10.2.5	Step-by-Step Example: Empirical Rule	37
10.3	Probability and Association: The Mechanics of Diversification	37
10.3.1	Pearson Correlation	38
10.3.2	Strategic Interpretation: Equity Cluster Risk	38
10.3.3	Probability Complement	38
10.3.4	Union or Addition Rule	38
10.3.5	Conditional Probability	38
10.3.6	Independence	39
10.3.7	Mutual Exclusivity	39
10.3.8	Step-by-Step Example: Directionality in Cross-Tabulation	39
10.4	Regression Modeling: Quantifying Predictive Power	39
10.4.1	Multiple Linear Regression	40
10.4.2	Ordinary Least Squares	40
10.4.3	Coefficient of Determination	40
10.4.4	Adjusted R^2	40
10.4.5	Logistic Regression	41
10.4.6	Strategic Interpretation: Fit versus Complexity	41
10.4.7	Step-by-Step Example: Predicting Fund Return	41
10.5	Clustering and Distance Measures: Unsupervised Segmentation	42
10.5.1	Euclidean Distance	42
10.5.2	Manhattan Distance	42
10.5.3	Jaccard Similarity	42
10.5.4	Calinski–Harabasz Index	43
10.5.5	Strategic Interpretation: Feature Scaling	43
10.5.6	Step-by-Step Example: K-Means Assignment	43
10.6	Model Evaluation Metrics: Validating Analytical Rigor	44
10.6.1	Precision	44
10.6.2	Recall	44
10.6.3	F1 Score	44
10.6.4	Mean Absolute Error	45
10.6.5	Root Mean Squared Error	45
10.6.6	Strategic Interpretation: Diagnostic RMSE	45
10.6.7	Step-by-Step Example: Fraud Detection	46
10.7	Final Strategic Principle: “Visualize First”	46
11	Capital Markets and Asset Pricing Formulas	47
11.1	Systematic Risk: Beta	47
11.2	Capital Asset Pricing Model (CAPM)	48
11.3	Actual Stock Return	48

11.4	Alpha: Mispricing Signal	49
11.5	Bid–Ask Spread	50
11.6	Option Payoffs at Expiry	50
11.6.1	Call Option	50
11.6.2	Put Option	51
11.7	Put–Call Parity	51
12	Fintech Credit Risk and Fraud Detection Formulas	52
12.1	Linear Regression Risk Score	52
12.2	Logistic Regression and Probability of Default	53
12.3	Differential Risk-Adjusted Pricing	53
13	Corporate Valuation Formulas	54
13.1	Free Cash Flow to the Firm (FCFF)	54
13.2	After-Tax Cost of Debt	55
13.3	Weighted Average Cost of Capital (WACC)	56
13.4	Constant-Growth Perpetuity and Terminal Value	56
14	Agentic AI Custom Scoring Logic	57
14.1	Mock Loan Officer Risk Score	57
15	Quick Formula Summary	59
16	Summary of Financial and Cryptocurrency Analytics Formulas	59
16.1	Daily Return of a Cryptocurrency	60
16.2	Purchasing Power Loss Due to Inflation	60
16.3	Sharpe Ratio	61
16.4	Capital Asset Pricing Model (CAPM)	62
16.5	Beta (β)	62
16.6	Alpha (α) or Abnormal Return	63
16.7	Gordon Growth Model	64
16.8	Collateralization Ratio	64
16.9	Summary Interpretation	65
17	Levels of Measurement: Nominal, Ordinal, Interval, and Ratio Data	65
17.1	What is Data?	65
17.2	The Four Levels of Measurement	66
17.2.1	Nominal Data	66
17.2.2	Ordinal Data	67
17.2.3	Interval Data	67
17.2.4	Ratio Data	68
17.3	Summary of the Four Measurement Levels	69
17.4	Important Statistical Terms	69
17.5	A Simple Example	70
17.6	Memory Trick: N–O–I–R	70
17.7	Categorical and Quantitative Data	70
17.7.1	Categorical Data	71
17.7.2	Quantitative Data	71
17.8	Complete Classification Diagram	71

17.9 Important Correction: Numbers Do Not Always Mean Quantitative Data	72
18 Cross-Tabulation and Categorical Data Analysis	72
18.1 Movie Cross-Tabulation	73
18.2 Interpreting Individual Cells	73
18.3 Checking the Row Totals	73
18.4 Checking the Column Totals	74
18.5 Understanding Crosstab Percentages	74
18.6 Strategic Interpretation	75
18.7 Conditional Probability from the Crosstab	75
18.8 How to Read Any Cross-Tabulation	76
19 Fundamental Types of Probability	76
19.1 Contingency Table	77
19.2 1. Marginal (Unconditional) Probability	77
19.3 2. Joint Probability	77
19.4 3. Conditional Probability	78
19.5 The “Shrinking Denominator” Concept	80
19.6 4. Union of Events	80
19.7 5. Complement of an Event	81
20 Key Differences Between Marginal, Joint, and Conditional Probability	82
21 A Simple Way to Remember the Three Concepts	82
22 Financial Interpretation	83
23 Important Exam Insight	84
24 Quick Revision Summary	84
25 Probability and Statistical Concepts	85
25.1 1. Marginal, Joint, and Conditional Probability	85
25.1.1 1.1 Marginal Probability	85
25.1.2 1.2 Joint Probability	85
25.1.3 1.3 Conditional Probability	86
25.1.4 1.4 The Shrinking Denominator	87
25.1.5 1.5 Comparison	87
25.2 2. Union of Events	87
25.3 3. Complement of an Event	88
25.4 4. Coefficient of Variation	88
25.4.1 4.1 Definition	88
25.4.2 4.2 Conceptual Meaning	89
25.4.3 4.3 BITSOM Equity Fund	89
25.4.4 4.4 BITSOM Debt Fund	90
25.4.5 4.5 Interpretation	90
25.5 5. Practice Question: Coefficient of Variation	90
25.6 6. Calinski–Harabasz Index	91
25.6.1 6.1 Definition	91
25.6.2 6.2 Between-Cluster Sum of Squares	91

25.6.3	6.3 Within-Cluster Sum of Squares	92
25.6.4	6.4 Practice Calculation	92
25.7	7. Z-Score	93
25.7.1	7.1 Definition	93
25.7.2	7.2 Conceptual Meaning	93
25.7.3	7.3 Practice Example	94
25.7.4	7.4 Outlier Detection	94
25.8	8. p-Value	95
25.8.1	8.1 Definition	95
25.8.2	8.2 Individual Regression Coefficient	95
25.8.3	8.3 Overall F-Test	96
25.8.4	8.4 Practice Example	96
25.8.5	8.5 Important Interpretation of p-Value	98
25.9	9. Final Exam Cheat Sheet	98
25.10	10. Key Exam Rules	98
26	Confusion Matrix Classification Evaluation Metrics	99
26.1	What is a Confusion Matrix?	99
26.2	The Four Core Cells	100
26.3	Accuracy	100
26.4	Recall (Sensitivity)	101
26.5	Precision	101
26.6	F1 Score	101
27	Precision–Recall Trade-off in Finance	101
27.1	Case A: The Risk-Averse Lender – Prioritizing Recall	102
27.2	Case B: The Growth-Focused Acquisition Desk – Prioritizing Precision	102
27.3	Case C: Real-Time Fraud Prevention – A Balanced Objective	103
28	Key Takeaway	103
29	Decentralized Autonomous Organizations (DAOs)	103
29.1	Core Characteristics of a DAO	103
29.2	How DAO Decisions and Rule Changes Work	104
29.3	DAO Governance vs. Traditional Governance	104
29.4	Key Risks and Challenges of DAOs	104
29.4.1	1. Whale Concentration	105
29.4.2	2. Voter Apathy	105
29.4.3	3. Legal and Regulatory Uncertainty	105
30	Key Takeaway	106

1 Time Value of Money (TVM) & Cash Flow Valuation

1.1 Future Value – One Period

Formula:

$$FV = PV(1 + r)$$

Example: You invest \$1,000 today at an annual interest rate of 5%. After one year:

$$FV = 1,000(1 + 0.05)$$

$$FV = \$1,050$$

1.2 Future Value – Multiple Periods

Formula:

$$FV = PV(1 + r)^t$$

Example: You invest \$1,000 at 5% interest for 2 years:

$$FV = 1,000(1 + 0.05)^2$$

$$FV = \$1,102.50$$

Historical example: If Manhattan's \$24 purchase price in 1626 were invested at 5% for 400 years:

$$FV = 24(1.05)^{400}$$

$$FV \approx \$7.17 \text{ billion}$$

1.3 Present Value – One Period

Formula:

$$PV = \frac{FV}{1 + r}$$

Example: To receive \$1,000 in exactly one year at a discount rate of 5%:

$$PV = \frac{1,000}{1 + 0.05}$$

$$PV = \$952.38$$

1.4 Present Value – Multiple Periods

Formula:

$$PV = \frac{FV}{(1+r)^t}$$

Example: To receive \$1,000 in two years at a discount rate of 5%:

$$PV = \frac{1,000}{(1+0.05)^2}$$

$$PV = \$907.03$$

1.5 Time to Reach a Future Value

Formula:

$$t = \frac{\log(FV/PV)}{\log(1+r)}$$

Example: To determine how long it takes to double your money at an interest rate of 5%:

$$\frac{FV}{PV} = 2$$

$$t = \frac{\log(2)}{\log(1.05)}$$

$$t \approx 14.21 \text{ years}$$

1.6 Present Value of an Annuity

Formula:

$$PV_0 = CF \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

Example: A vacation house generates a constant net rental stream of \$30,000 per year for 4 years at a discount rate of 20%:

$$PV_0 = 30,000 \left[\frac{1 - (1.20)^{-4}}{0.20} \right]$$

$$PV_0 = \$77,662.04$$

1.7 Present Value of a Perpetuity

Formula:

$$PV_0 = \frac{CF_1}{r}$$

Example: A toll road generates a steady income of 10 crore per year forever. The required rate of return is 10%:

$$PV_0 = \frac{10 \text{ crore}}{0.10}$$

$$PV_0 = 100 \text{ crore}$$

1.8 Present Value of a Growing Perpetuity

Formula:

$$PV_0 = \frac{CF_1}{r - g}$$

where

$$r > g$$

Example: A firm is expected to pay next year's free cash flow of \$50 million, which grows at 5% forever. The cost of capital is 10%:

$$PV_0 = \frac{50 \text{ million}}{0.10 - 0.05}$$

$$PV_0 = \$1,000 \text{ million}$$

2 Compounding & Interest Rates

2.1 Fisher Equation

Exact Formula:

$$1 + r_{\text{nominal}} = (1 + r_{\text{real}})(1 + i)$$

Rearranged for the Real Rate:

$$r_{\text{real}} = \frac{1 + r_{\text{nominal}}}{1 + i} - 1$$

Example: A nominal return of 8% with inflation of 3%:

$$r_{\text{real}} = \frac{1.08}{1.03} - 1$$

$$r_{\text{real}} \approx 4.854\%$$

The approximate relationship is:

$$r_{\text{real}} \approx r_{\text{nominal}} - i$$

Thus,

$$8\% - 3\% = 5\%$$

This additive approximation is not exact.

2.2 APR vs. EAR

APR Formula:

$$APR = (\text{rate per period}) \times (\text{periods per year})$$

EAR Formula:

$$EAR = \left(1 + \frac{APR}{m}\right)^m - 1$$

where m is the number of compounding periods per year.

Example: A loan charges 1% per month:

$$APR = 1\% \times 12$$

$$\boxed{APR = 12\%}$$

With quarterly compounding, $m = 4$:

$$EAR = \left(1 + \frac{0.12}{4}\right)^4 - 1$$

$$EAR = (1.03)^4 - 1$$

$$\boxed{EAR \approx 12.55\%}$$

2.3 Continuous Compounding

Effective Annual Rate:

$$EAR = e^{APR} - 1$$

Future Value:

$$FV = PV e^{rt}$$

Example: Invest \$1,000 at a continuously compounded rate of 5% for 10 years:

$$FV = 1,000e^{0.05(10)}$$

$$FV = 1,000e^{0.5}$$

$$\boxed{FV \approx \$1,648.72}$$

3 Risk, Return & Basic Statistics

3.1 Expected Return

Formula:

$$\mathbb{E}(R) = \sum_{i=1}^n P_i R_i$$

Example:

$$\mathbb{E}(R) = (0.25)(-6\%) + (0.50)(10\%) + (0.25)(22\%)$$

$$\mathbb{E}(R) = -1.5\% + 5\% + 5.5\%$$

$$\boxed{\mathbb{E}(R) = 9\%}$$

3.2 Variance and Standard Deviation

Variance:

$$\sigma^2 = \sum_i P_i (R_i - \mathbb{E}(R))^2$$

Standard Deviation:

$$\sigma = \sqrt{\sigma^2}$$

Using $\mathbb{E}(R) = 9\%$:

$$\begin{aligned}\sigma^2 &= 0.25(-6\% - 9\%)^2 \\ &\quad + 0.50(10\% - 9\%)^2 \\ &\quad + 0.25(22\% - 9\%)^2\end{aligned}$$

$$\sigma^2 = 0.25(225) + 0.50(1) + 0.25(169)$$

$$\sigma^2 = 56.25 + 0.50 + 42.25$$

$$\boxed{\sigma^2 = 99}$$

Therefore,

$$\sigma = \sqrt{99}$$

$$\boxed{\sigma \approx 9.95\%}$$

3.3 Historical Average Return

Formula:

$$\bar{R} = \frac{1}{T} \sum_{t=1}^T R_t$$

Example: Returns over four years are 10%, 12%, 8%, 15%:

$$\bar{R} = \frac{10\% + 12\% + 8\% + 15\%}{4}$$

$$\boxed{\bar{R} = 11.25\%}$$

3.4 Historical Variance – Sample Variance

Formula:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Example: Returns are 5%, 7%, 9%, with $\bar{R} = 7\%$:

$$\text{Var}(R) = \frac{(5\% - 7\%)^2 + (7\% - 7\%)^2 + (9\% - 7\%)^2}{3 - 1}$$

$$= \frac{4 + 0 + 4}{2}$$

$$\boxed{\text{Var}(R) = 4\%^2}$$

Therefore,

$$\sigma = \sqrt{4}$$

$$\boxed{\sigma = 2\%}$$

3.5 Covariance and Correlation

Covariance:

$$\text{Cov}(R_A, R_B) = \frac{1}{T-1} \sum_{t=1}^T (R_{At} - \bar{R}_A)(R_{Bt} - \bar{R}_B)$$

Correlation:

$$\rho_{AB} = \frac{\text{Cov}(R_A, R_B)}{\sigma_A \sigma_B}$$

Example:

$$\sigma_A = 10\%, \quad \sigma_B = 15\%, \quad \text{Cov}(R_A, R_B) = 0.012$$

$$\rho_{AB} = \frac{0.012}{(0.10)(0.15)}$$

$$\rho_{AB} = 0.80$$

This represents a strongly positive correlation.

4 Portfolio Theory & Diversification

4.1 Portfolio Return

Formula:

$$\mathbb{E}(R_p) = w_A \mathbb{E}(R_A) + w_B \mathbb{E}(R_B)$$

Example:

$$w_A = 40\%, \quad \mathbb{E}(R_A) = 12\%$$

$$w_B = 60\%, \quad \mathbb{E}(R_B) = 8\%$$

$$\mathbb{E}(R_p) = (0.40)(12\%) + (0.60)(8\%)$$

$$= 4.8\% + 4.8\%$$

$$\mathbb{E}(R_p) = 9.6\%$$

4.2 Portfolio Variance – Two Assets

Formula:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Example:

$$w_X = 60\%, \quad w_Y = 40\%$$

$$\sigma_X = 20\%, \quad \sigma_Y = 15\%$$

$$\rho_{XY} = 0.30$$

Therefore,

$$\begin{aligned} \sigma_p^2 &= (0.60)^2(0.20)^2 + (0.40)^2(0.15)^2 \\ &\quad + 2(0.60)(0.40)(0.30)(0.20)(0.15) \end{aligned}$$

$$\sigma_p^2 = 0.0144 + 0.0036 + 0.00432$$

$$\sigma_p^2 = 0.02232$$

Thus,

$$\sigma_p = \sqrt{0.02232}$$

$$\sigma_p \approx 14.94\%$$

The weighted-average volatility is:

$$(0.60)(20\%) + (0.40)(15\%) = 18\%$$

Since

$$14.94\% < 18\%$$

diversification reduces portfolio risk when

$$\rho < 1.$$

4.3 Variance of an Equally Weighted Portfolio

Formula:

$$\sigma_p^2 = \frac{1}{N}\bar{\sigma}^2 + \left(1 - \frac{1}{N}\right)\overline{\text{Cov}}$$

Example:

$$N = 20, \quad \bar{\sigma}^2 = 0.09, \quad \overline{\text{Cov}} = 0.02$$

$$\begin{aligned}\sigma_p^2 &= \frac{0.09}{20} + \left(1 - \frac{1}{20}\right)(0.02) \\ &= 0.0045 + 0.019\end{aligned}$$

$$\sigma_p^2 = 0.0235$$

As

$$N \rightarrow \infty,$$

the first term approaches zero:

$$\frac{1}{N}\bar{\sigma}^2 \rightarrow 0$$

Therefore, idiosyncratic risk vanishes while systematic risk remains.

5 Asset Pricing, CAPM & Risk Metrics

5.1 Beta

Formula:

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\sigma_M^2}$$

Example:

$$\text{Cov}(R_i, R_M) = 0.03, \quad \sigma_M^2 = 0.02$$

$$\beta_i = \frac{0.03}{0.02}$$

$$\boxed{\beta_i = 1.50}$$

A beta of 1.50 indicates that the stock has greater market sensitivity than the market portfolio.

5.2 CAPM – Expected Return

Formula:

$$\mathbb{E}(R_i) = r_f + \beta_i [\mathbb{E}(R_M) - r_f]$$

Example:

$$r_f = 5\%, \quad \mathbb{E}(R_M) = 12\%, \quad \beta_i = 1.5$$

$$\mathbb{E}(R_i) = 5\% + 1.5(12\% - 5\%)$$

$$= 5\% + 10.5\%$$

$$\boxed{\mathbb{E}(R_i) = 15.5\%}$$

5.3 Alpha

Formula:

$$\alpha_i = R_{\text{actual}} - \mathbb{E}(R_{\text{CAPM}})$$

Using the CAPM expected return of 15.5%:

If the actual return is 20%:

$$\alpha = 20\% - 15.5\%$$

$$\boxed{\alpha = +4.5\%}$$

If the actual return is 12%:

$$\alpha = 12\% - 15.5\%$$

$$\boxed{\alpha = -3.5\%}$$

5.4 Portfolio Beta

Formula:

$$\beta_p = \sum_i w_i \beta_i$$

Example:

$$w_A = 70\%, \quad \beta_A = 1.2$$

$$w_B = 30\%, \quad \beta_B = -0.5$$

$$\begin{aligned}\beta_p &= (0.70)(1.2) + (0.30)(-0.5) \\ &= 0.84 - 0.15 \\ &= \boxed{0.69}\end{aligned}$$

5.5 Sharpe Ratio

Formula:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

Example:

$$R_p = 12\%, \quad R_f = 4\%, \quad \sigma_p = 10\%$$

$$\text{Sharpe Ratio} = \frac{12\% - 4\%}{10\%}$$

$$\boxed{\text{Sharpe Ratio} = 0.80}$$

6 Market Microstructure & Execution Cost

6.1 Bid-Ask Spread and Mid-Price

Spread:

$$\text{Spread} = \text{Ask Price} - \text{Bid Price}$$

Mid-Price:

$$\text{Mid-Price} = \frac{\text{Best Ask} + \text{Best Bid}}{2}$$

Example:

$$\text{Ask} = \$100.05, \quad \text{Bid} = \$99.95$$

$$\text{Spread} = 100.05 - 99.95$$

$$\boxed{\text{Spread} = \$0.10}$$

$$\text{Mid-Price} = \frac{100.05 + 99.95}{2}$$

$$\boxed{\text{Mid-Price} = \$100.00}$$

6.2 Percentage Spread

Formula:

$$\text{Percentage Spread} = \frac{\text{Ask Price} - \text{Bid Price}}{\text{Mid-Price}}$$

Example:

$$\text{Percentage Spread} = \frac{0.10}{100.00}$$

$$= 0.001$$

Therefore,

$$\boxed{\text{Percentage Spread} = 0.10\%}$$

6.3 Volume-Weighted Average Price (VWAP)

Formula:

$$VWAP = \frac{\sum_i (P_i \times V_i)}{\sum_i V_i}$$

Example: Execute an order in two tranches:

100 shares at \$50

and

300 shares at \$52

Then,

$$\begin{aligned} VWAP &= \frac{(50)(100) + (52)(300)}{100 + 300} \\ &= \frac{5,000 + 15,600}{400} \end{aligned}$$

$$\boxed{VWAP = \$51.50}$$

7 Corporate Valuation & Free Cash Flow

7.1 Free Cash Flow to the Firm

Formula:

$$FCF = EBIT - Taxes + Depreciation - CapEx - \Delta NWC$$

Alternative Formula Starting with Net Income:

$$FCF = Net\ Income + Depreciation - CapEx - \Delta NWC$$

Example:

Given:

$$EBIT = \$300 \text{ million}$$

$$Taxes = \$90 \text{ million}$$

$$Depreciation = \$50 \text{ million}$$

$$CapEx = \$120 \text{ million}$$

$$\Delta NWC = \$10 \text{ million}$$

Therefore,

$$\begin{aligned} FCF &= 300 - 90 + 50 - 120 - 10 \\ &= \boxed{\$130 \text{ million}} \end{aligned}$$

8 Summary Tips for Exams

8.1 Discounting Treatment

Always match nominal cash flows with nominal discount rates:

$$\boxed{\text{Nominal Cash Flows} \longleftrightarrow \text{Nominal Discount Rates}}$$

Similarly, match real cash flows with real discount rates:

$$\boxed{\text{Real Cash Flows} \longleftrightarrow \text{Real Discount Rates}}$$

8.2 Cash Flow Timelines

When using DCF shortcuts such as annuities and perpetuities, remember that the formula calculates the value one period before the first cash flow arrives.

For example:

$$PV_0 = \frac{CF_1}{r}$$

where CF_1 is the cash flow received at $t = 1$.

8.3 Risk Diversification

In a highly diversified portfolio, firm-specific (idiosyncratic) risk approaches zero:

$$\boxed{\text{Idiosyncratic Risk} \rightarrow 0}$$

while systematic risk remains:

$$\boxed{\text{Systematic Risk remains}}$$

Under CAPM, systematic risk is the risk that is priced in equilibrium.

$$\boxed{\text{Diversification eliminates idiosyncratic risk, not systematic risk.}}$$

Quick Formula Reference

$$\alpha_i = R_{\text{actual}} - \mathbb{E}(R_{\text{CAPM}})$$

$$\beta_p = \sum_i w_i \beta_i$$

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

$$\text{Spread} = \text{Ask} - \text{Bid}$$

$$\text{Mid-Price} = \frac{\text{Ask} + \text{Bid}}{2}$$

$$VWAP = \frac{\sum_i P_i V_i}{\sum_i V_i}$$

$$FCF = EBIT - \text{Taxes} + \text{Depreciation} - \text{CapEx} - \Delta NWC$$

$$FV = PV(1+r)^t \quad PV = \frac{FV}{(1+r)^t} \quad t = \frac{\log(FV/PV)}{\log(1+r)} \quad PV_{\text{annuity}} = CF \left[\frac{1 - (1+r)^{-n}}{r} \right] \quad PV_{\text{perpetuity}} = \frac{CF}{r}$$

$$r_f + \beta_i[\mathbb{E}(R_M) - r_f]$$

$$\alpha_i = R_{\text{actual}} - \mathbb{E}(R_{\text{CAPM}})$$

$$\beta_p = \sum_i w_i \beta_i$$

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

$$\text{Spread} = \text{Ask} - \text{Bid}$$

$$\text{Mid-Price} = \frac{\text{Ask} + \text{Bid}}{2}$$

$$\text{VWAP} = \frac{\sum_i P_i V_i}{\sum_i V_i}$$

$$\text{FCF} = \text{EBIT} - \text{Taxes} + \text{Depreciation} - \text{CapEx} - \Delta \text{NWC}$$

9 Fintech Economics, Credit Risk & Model Evaluation

This section introduces several important quantitative tools used in fintech and credit-risk management. These measures help firms evaluate customer profitability, estimate default risk, assess model accuracy, separate good and bad borrowers, and determine profitable lending cut-offs.

9.1 Customer Acquisition Cost (CAC) and Lifetime Value (LTV)

9.1.1 Concept

Customer Acquisition Cost (CAC) measures the average amount a firm spends to acquire one new customer. Lifetime Value (LTV), in contrast, measures the economic value that a customer is expected to generate over the entire customer relationship.

The comparison between CAC and LTV is fundamental to understanding whether a fintech firm's growth is economically sustainable.

9.1.2 Key Formulas

Customer Acquisition Cost:

$$\text{CAC} = \frac{\text{Total Acquisition Spending}}{\text{Total New Users Acquired}}$$

This formula calculates the average cost of acquiring one new customer.

Net Profit per User:

$$\text{Net Profit per User} = \text{LTV} - \text{CAC}$$

This measures the economic contribution of an average customer after recovering the acquisition cost.

Profitability Rule:

$$\text{LTV} > \text{CAC}$$

If LTV exceeds CAC, each new customer is expected to create positive economic value. Conversely,

$$\boxed{LTV < CAC \implies \text{Value Destruction}}$$

If CAC exceeds LTV, acquiring additional customers increases the firm's losses rather than creating value.

9.1.3 Worked Example

Suppose a Buy Now, Pay Later (BNPL) fintech spends 100 crore on digital advertising, cashback incentives, and onboarding. The firm acquires 1 million new users.

The customer acquisition cost is:

$$CAC = \frac{100 \text{ crore}}{1 \text{ million users}}$$

Since

$$1 \text{ crore} = 10,000,000,$$

we have:

$$100 \text{ crore} = 1,000,000,000$$

Therefore,

$$CAC = \frac{1,000,000,000}{1,000,000}$$

$$\boxed{CAC = 1,000 \text{ per user}}$$

Scenario A: Healthy Growth Suppose the average customer repeatedly uses the BNPL platform and generates an LTV of 3,000.

$$\text{Net Profit per User} = LTV - CAC$$

$$= 3,000 - 1,000$$

$$\boxed{\text{Net Profit per User} = +2,000}$$

Since

$$3,000 > 1,000,$$

the customer generates positive economic value.

Scenario B: Value Destruction Suppose most users abandon the platform after only a few low-margin transactions, resulting in an LTV of 800.

$$\text{Net Profit per User} = 800 - 1,000$$

$$\boxed{\text{Net Profit per User} = -200}$$

Here,

$$800 < 1,000.$$

Therefore, the firm loses 200 for every acquired customer, before considering other operating costs.

Interpretation The key relationship is:

$$\boxed{\text{LTV} > \text{CAC} \Rightarrow \text{Potentially Sustainable Growth}}$$

whereas

$$\boxed{\text{LTV} < \text{CAC} \Rightarrow \text{Customer Acquisition Destroys Value}}$$

9.2 Logistic Regression – The Baseline Underwriting Model

9.2.1 Concept

Logistic regression is widely used in credit-risk modelling because the dependent variable is often binary. For example,

$$Y_i = \begin{cases} 1, & \text{if borrower } i \text{ defaults,} \\ 0, & \text{if borrower } i \text{ performs.} \end{cases}$$

The model converts borrower characteristics into a probability of default between 0 and 1.

9.2.2 Logit Form

The relationship between the probability of default and borrower characteristics can be written as:

$$\boxed{\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_n X_{ni}}$$

where π_i represents the probability that borrower i defaults.

The term

$$\frac{\pi_i}{1 - \pi_i}$$

represents the **odds of default**, while

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right)$$

is called the **log-odds** or **logit**.

9.2.3 Probability Form

Define the linear predictor:

$$\eta_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_n X_{ni}$$

The probability of default is then:

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$$

or equivalently,

$$\pi_i = \frac{1}{1 + e^{-\eta_i}}$$

This transformation ensures that:

$$0 < \pi_i < 1.$$

9.2.4 Meaning of the Variables

- π_i : Probability that borrower i defaults.
- Y_i : Binary default outcome.
- $X_{1i}, X_{2i}, \dots, X_{ni}$: Borrower characteristics or predictors.
- β_0 : Intercept of the model.
- $\beta_1, \beta_2, \dots, \beta_n$: Regression coefficients estimated using historical data.
- η_i : Linear predictor or score before applying the logistic transformation.

Typical predictors can include income, credit score, debt level, repayment history, transaction behaviour, and other legally and appropriately collected risk variables.

9.2.5 Worked Example

Suppose a simplified underwriting model uses only one predictor: the applicant's bank balance X .

Assume the estimated coefficients are:

$$\beta_0 = -10.65$$

and

$$\beta_1 = 0.0055.$$

For an applicant with a bank balance of

$$X = 2000,$$

the linear predictor is:

$$\begin{aligned}\eta &= \beta_0 + \beta_1 X \\ &= -10.65 + (0.0055)(2000) \\ &= -10.65 + 11\end{aligned}$$

$$\boxed{\eta = 0.35}$$

We now transform the log-odds into a probability:

$$\pi = \frac{e^{0.35}}{1 + e^{0.35}}$$

Since

$$e^{0.35} \approx 1.419,$$

we obtain:

$$\begin{aligned}\pi &= \frac{1.419}{1 + 1.419} \\ &= \frac{1.419}{2.419}\end{aligned}$$

$$\boxed{\pi \approx 0.5866 \approx 58.7\%}$$

Thus, the model estimates approximately a 59% probability of default. If the lender uses a default-probability threshold of 50%, then:

$$\pi > 0.50$$

and the application would be classified as high risk under this simplified decision rule.

$$\boxed{\text{Decision: Decline}}$$

Interpretation Logistic regression does not directly say that a borrower will definitely default. Instead, it estimates a probability:

$$\boxed{\pi_i = P(Y_i = 1 \mid X_i)}$$

The lender then compares this probability with a chosen decision threshold.

9.3 Confusion Matrix Evaluation: Sensitivity and Specificity

9.3.1 Concept

A confusion matrix compares the model's predictions with the actual outcomes. It is particularly useful when evaluating whether a credit model correctly identifies defaulters and performing borrowers.

9.3.2 Four Possible Outcomes

- **True Positive (TP):** A defaulter is correctly classified as a defaulter.
- **False Negative (FN):** A defaulter is incorrectly classified as a good borrower.
- **True Negative (TN):** A performing borrower is correctly classified as good.
- **False Positive (FP):** A performing borrower is incorrectly classified as a defaulter.

9.3.3 Sensitivity

Sensitivity measures the model's ability to identify actual defaulters.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

A high sensitivity means that the model catches a large proportion of actual bad borrowers.

9.3.4 Specificity

Specificity measures the model's ability to correctly identify performing borrowers.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

A high specificity means that the model correctly recognizes a large proportion of good borrowers.

9.3.5 Worked Example

Suppose a bank evaluates a portfolio containing:

100 actual defaulters

and

1,000 actual good borrowers.

For the defaulters:

$$TP = 85, \quad FN = 15.$$

Therefore,

$$\begin{aligned} \text{Sensitivity} &= \frac{85}{85 + 15} \\ &= \frac{85}{100} \end{aligned}$$

$$\text{Sensitivity} = 85\%$$

Thus, the model correctly identifies 85% of actual defaulters.

For the good borrowers:

$$TN = 900, \quad FP = 100.$$

Therefore,

$$\begin{aligned} \text{Specificity} &= \frac{900}{900 + 100} \\ &= \frac{900}{1000} \end{aligned}$$

$$\boxed{\text{Specificity} = 90\%}$$

Thus, the model correctly identifies 90% of performing borrowers.

Interpretation The model has:

$$\boxed{\text{Sensitivity} = 85\%}$$

and

$$\boxed{\text{Specificity} = 90\%}.$$

This means that it catches 85% of defaulters while correctly recognizing 90% of good borrowers.

9.4 Kolmogorov–Smirnov (KS) Statistic

9.4.1 Concept

The Kolmogorov–Smirnov (KS) statistic measures how effectively a credit-risk model separates good borrowers from bad borrowers.

The borrowers are usually ranked according to their risk scores and divided into score ranges or bands.

9.4.2 KS Distance

At score range j , the KS distance is:

$$\boxed{\text{KS Distance}_j = |\text{Cumulative \% of Bads Captured}_j - \text{Cumulative \% of Goods Captured}_j|}$$

The overall KS statistic is the maximum distance:

$$\boxed{\text{KS} = \max_j (\text{KS Distance}_j)}$$

The higher the KS statistic, the greater the separation between the distributions of good and bad borrowers.

9.4.3 Worked Example

Suppose there are 100 borrowers divided into five score ranges, with 20 borrowers in each range.

The portfolio contains:

80 good borrowers

and

20 bad borrowers.

Score Range 1 Suppose the worst score range contains:

8 bad borrowers

and

12 good borrowers.

The cumulative percentage of bad borrowers captured is:

$$\frac{8}{20} \times 100 = 40\%.$$

The cumulative percentage of good borrowers captured is:

$$\frac{12}{80} \times 100 = 15\%.$$

Therefore,

$$KS_1 = |40\% - 15\%|$$

$$\boxed{KS_1 = 25\% = 0.25}$$

Score Range 2 Suppose the second range contains another 6 bad borrowers and 14 good borrowers.

Cumulative bad borrowers:

$$8 + 6 = 14.$$

Therefore,

$$\frac{14}{20} \times 100 = 70\%.$$

Cumulative good borrowers:

$$12 + 14 = 26.$$

Therefore,

$$\frac{26}{80} \times 100 = 32.5\%.$$

Thus,

$$KS_2 = |70\% - 32.5\%|$$

$$KS_2 = 37.5\% \approx 0.375$$

Score Range 3 Suppose the third range contains 3 additional bad borrowers and 17 additional good borrowers.

Cumulative bad borrowers:

$$14 + 3 = 17.$$

Therefore,

$$\frac{17}{20} \times 100 = 85\%.$$

Cumulative good borrowers:

$$26 + 17 = 43.$$

Therefore,

$$\frac{43}{80} \times 100 = 53.75\%.$$

Hence,

$$KS_3 = |85\% - 53.75\%|$$

$$KS_3 = 31.25\% \approx 0.313$$

Since

$$37.5\% > 31.25\% > 25\%,$$

the maximum separation occurs at Score Range 2.

Therefore,

$$KS \approx 37.5\% = 0.375$$

Interpretation A higher KS value indicates better separation between good and bad borrowers. In this example, the model concentrates a large proportion of bad borrowers toward the worst-scoring end of the portfolio.

9.5 Expected Loss Minimization and Score Cut-Offs

9.5.1 Concept

A credit model does not need to achieve perfect classification to be economically useful. The lender must ultimately decide which applications to approve and which to reject.

The optimal cut-off depends on the revenue earned from a good loan and the expected loss from a bad loan.

9.5.2 Net Profit Formula

For a particular score range:

$$\text{Net Profit per Range} = (\text{Good Accounts} \times \text{Revenue per Good Loan}) - (\text{Bad Accounts} \times \text{Loss per Bad Loan})$$

The basic decision rule is:

$$\text{Approve if Net Profit per Range} > 0$$

and

$$\text{Decline if Net Profit per Range} < 0$$

This approach links statistical model performance directly to economic outcomes.

9.5.3 Worked Example

Suppose a lender earns:

\$100

for every performing loan and loses:

\$200

for every defaulting loan.

Score Range 5 – Best Range Suppose the range contains:

19 good accounts

and

1 bad account.

The net profit is:

$$\begin{aligned}\text{Net Profit}_5 &= (19 \times \$100) - (1 \times \$200) \\ &= \$1,900 - \$200 \\ &= \boxed{+\$1,700}\end{aligned}$$

Since

$$+\$1,700 > 0,$$

the range should be approved under this simplified rule.

$$\boxed{\text{Decision: Approve}}$$

Score Range 1 – Worst Range Suppose the range contains:

12 good accounts

and

8 bad accounts.

Then:

$$\begin{aligned}\text{Net Profit}_1 &= (12 \times \$100) - (8 \times \$200) \\ &= \$1,200 - \$1,600 \\ &= \boxed{-\$400}\end{aligned}$$

Since

$$-\$400 < 0,$$

the lender should reject this range under the simplified decision rule.

Decision: Decline

Interpretation Notice that good borrowers outnumber bad borrowers in Range 1:

$$12 > 8.$$

Nevertheless, the range is unprofitable because the loss associated with each bad loan is twice the revenue generated by each good loan.

Thus, the optimal credit cut-off should be based not only on classification accuracy but also on the economic consequences of different errors.

9.6 Peer-to-Peer (P2P) Lending Platform Calculations

9.6.1 Concept

P2P lending platforms connect borrowers with lenders. The platform may charge a spread or fee for facilitating the transaction.

The following calculations determine the gross interest generated, the platform's spread, and the lender's net return.

9.6.2 Interest Earned

$$\boxed{\text{Interest Earned} = \text{Principal} \times \text{Annual Interest Rate}}$$

This formula calculates the gross interest generated by the loan for one year, assuming a simple-interest framework.

9.6.3 Platform Spread

$$\boxed{\text{Platform Spread} = \text{Principal} \times \text{Platform Spread Percentage}}$$

The platform spread represents the amount retained by the intermediary under the simplified assumptions of this example.

9.6.4 Lender Net Return

$$\boxed{\text{Lender Net Return} = \text{Interest Earned} - \text{Platform Spread}}$$

Thus, the lender's net return is the gross interest received after subtracting the platform's spread.

9.6.5 Worked Example

Suppose a borrower receives a:

$$100$$

loan at an annual interest rate of:

$$25\%.$$

The gross interest earned is:

$$\text{Interest Earned} = 100 \times 25\%$$

$$\boxed{\text{Interest Earned} = 25}$$

Suppose the P2P platform charges a 3% spread on the principal.

Then:

$$\text{Platform Spread} = 100 \times 3\%$$

$$\boxed{\text{Platform Spread} = 3}$$

Therefore, the lender's net return is:

$$\text{Lender Net Return} = 25 - 3$$

$$\boxed{\text{Lender Net Return} = 22}$$

Interpretation Under the simplified assumptions, the borrower pays 25 in annual interest, the platform retains 3 as its spread, and 22 remains as the lender's net interest return.

9.7 Quick Formula Reference

$$\text{CAC} = \frac{\text{Total Acquisition Spending}}{\text{Total New Users Acquired}}$$

$$\text{Net Profit per User} = \text{LTV} - \text{CAC}$$

$$\text{LTV} > \text{CAC} \quad (\text{Positive Unit Economics})$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \sum_{k=1}^n \beta_k X_{ki}$$

$$\eta_i = \beta_0 + \sum_{k=1}^n \beta_k X_{ki}$$

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

$$\text{KS}_j = |\text{Cumulative \% Bads}_j - \text{Cumulative \% Goods}_j|$$

$$\text{KS} = \max_j(\text{KS}_j)$$

$$\begin{aligned} \text{Net Profit} &= (\text{Good Accounts} \times \text{Revenue per Good Loan}) \\ &\quad - (\text{Bad Accounts} \times \text{Loss per Bad Loan}) \end{aligned}$$

$$\text{Interest Earned} = \text{Principal} \times \text{Annual Interest Rate}$$

$$\text{Platform Spread} = \text{Principal} \times \text{Platform Spread Percentage}$$

$$\text{Lender Net Return} = \text{Interest Earned} - \text{Platform Spread}$$

9.8 Key Takeaways

1. **CAC vs. LTV:** Sustainable fintech growth generally requires customer lifetime value to exceed customer acquisition cost.
2. **Logistic Regression:** Converts borrower characteristics into a probability of default between 0 and 1.
3. **Sensitivity:** Measures how effectively a model identifies actual defaulters.
4. **Specificity:** Measures how effectively a model identifies actual performing borrowers.
5. **KS Statistic:** Measures the maximum separation between cumulative bad and good distributions across risk-score ranges.

6. **Score Cut-Offs:** The optimal lending threshold should consider the economic cost of false approvals and false rejections, rather than relying only on statistical accuracy.
7. **P2P Economics:** Gross interest must be adjusted for platform spreads or fees to determine the lender's net return.

10 Financial Analytics: Formula Sheet and Strategic Reference Guide

10.1 Descriptive Statistics: Measuring Central Tendency and Dispersion

Descriptive statistics form the foundation of financial analysis. They allow analysts to summarize large datasets, such as multi-year daily fund returns, into a small number of meaningful statistical measures. These measures help distinguish the underlying performance of an investment from the short-term noise present in financial markets.

10.1.1 Arithmetic Mean

The arithmetic mean represents the average value of a dataset. In finance, it is commonly used to calculate the average return of an asset over a given period.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where:

- $\bar{x}$ = arithmetic mean,
- x_i = observation i ,
- n = total number of observations.

Strategic Use: The arithmetic mean provides a simple estimate of the typical return over a specific period. However, it can be heavily influenced by extreme observations.

10.1.2 Weighted Mean

A weighted mean assigns different levels of importance to different observations. It is particularly useful in portfolio analysis, where each asset has a different portfolio weight.

$$\bar{x}_w = \sum_{i=1}^n w_i x_i$$

where w_i represents the weight assigned to observation i .

Strategic Use: The weighted mean is used to calculate the expected return of a portfolio containing multiple assets with different allocations.

10.1.3 Median

The median is the middle observation after arranging the data in ascending or descending order. For an odd number of observations, the position of the median is:

$$\text{Median Position} = \frac{n + 1}{2}$$

Strategic Use: The median is more robust than the arithmetic mean when the dataset contains extreme values or outliers.

10.1.4 Percentiles and Quartiles

A percentile indicates the value below which a specified percentage of observations fall.

$$P_k = \frac{k}{100}(n + 1)$$

where k represents the desired percentile.

The most commonly used quartiles are:

$$Q_1 = P_{25}, \quad Q_2 = P_{50}, \quad Q_3 = P_{75}$$

Here, Q_2 is the median.

Strategic Use: Quartiles divide the dataset into four sections and help analysts understand the distribution of investment performance.

10.1.5 Range

The range measures the absolute difference between the maximum and minimum observations.

$$\text{Range} = x_{\max} - x_{\min}$$

Strategic Use: It provides a quick measure of the total spread in returns, although it is highly sensitive to extreme observations.

10.1.6 Interquartile Range

The interquartile range measures the spread of the middle 50% of observations.

$$IQR = Q_3 - Q_1$$

Strategic Use: Unlike the range, the IQR is relatively resistant to extreme values and therefore provides a better measure of typical dispersion.

10.1.7 Variance

Variance measures the average squared deviation of observations from their mean.

For a population:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Strategic Use: Variance quantifies the absolute dispersion of financial returns. Since deviations are squared, the resulting unit is squared as well.

10.1.8 Standard Deviation

Standard deviation is the square root of variance and is one of the most widely used measures of financial risk.

$$\sigma = \sqrt{\sigma^2}$$

Strategic Use: In finance, standard deviation is commonly interpreted as volatility. A higher standard deviation indicates greater variability in returns.

10.1.9 Coefficient of Variation

The coefficient of variation measures risk relative to the average return.

$$CV = \frac{\sigma}{\bar{x}} \times 100\%$$

Strategic Use: CV is useful when comparing investments that have different average return levels. A lower CV generally indicates less risk per unit of expected return.

10.1.10 Strategic Interpretation: Risk versus Stability

Average returns can sometimes be misleading. For example, suppose the Metals and FMCG sectors both generate an annual average return of 3.5%. Their risk profiles may nevertheless be very different.

If FMCG returns generally lie between 0% and 7%, while Metals returns range from -12% to +15%, the two sectors have the same average return but very different levels of volatility.

Similarly, if the Bitsom Equity Fund has an average return of approximately 1.88% and a coefficient of variation of approximately 123%, the fund exhibits substantial volatility relative to its average return.

Therefore, analysts should consider measures such as standard deviation, CV, and IQR rather than relying solely on the arithmetic mean.

10.1.11 Step-by-Step Example: Weighted Portfolio Return

Consider a portfolio with the following monthly returns and portfolio weights:

- Equity Fund: 1.9% return with a 60% weight,
- Debt Fund: 0.5% return with a 30% weight,
- Gold ETF: 0.8% return with a 10% weight.

The weighted portfolio return is:

$$\begin{aligned}\bar{x}_w &= (0.60)(1.9\%) + (0.30)(0.5\%) + (0.10)(0.8\%) \\ &= 1.14\% + 0.15\% + 0.08\% \\ &= \boxed{1.37\%}\end{aligned}$$

Thus, the expected monthly portfolio return is 1.37%.

While measures of central tendency describe the expected or typical outcome, the shape of the distribution provides additional information about potential tail risk.

10.2 Distributions and Shape: The Geometry of Risk

Financial models often rely on assumptions about the underlying distribution of returns. However, actual financial returns may exhibit skewness, fat tails, and extreme observations that differ substantially from a theoretical normal distribution.

10.2.1 Z-Score

The Z-score measures how many standard deviations an observation is away from the mean.

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = observed value,
- μ = population mean,
- σ = population standard deviation.

A commonly used rule is that observations with

$$|z| > 3$$

may be considered unusual or potential outliers under a normal-distribution assumption.

10.2.2 Empirical Rule

For a normally distributed variable:

$$\begin{array}{l} 68\% \text{ of observations lie within } \pm 1\sigma, \\ 95\% \text{ of observations lie within } \pm 2\sigma, \\ 99.7\% \text{ of observations lie within } \pm 3\sigma. \end{array}$$

This is commonly known as the 68-95-99.7 rule.

Strategic Use: The empirical rule helps analysts identify whether observed returns are unusually high or low relative to a normal-distribution benchmark.

10.2.3 Skewness

Skewness measures the asymmetry of a probability distribution.

Positive Skewness A positively skewed distribution has a longer right tail. It may be associated with frequent small losses and occasional very large gains.

$$\text{Mean} > \text{Median}$$

Negative Skewness A negatively skewed distribution has a longer left tail. It may involve frequent small gains but occasional large losses.

$$\boxed{\text{Mean} < \text{Median}}$$

10.2.4 Strategic Interpretation: Downside Risk

Suppose the Bitsom Equity Fund has:

$$\mu = 1.88\%, \quad \text{Median} = 2.25\%$$

Since

$$1.88\% < 2.25\%,$$

the distribution is negatively skewed. This may indicate greater exposure to large downside events.

In contrast, the Bitsom Debt Fund has:

$$\mu = 0.58\%, \quad \text{Median} = 0.50\%,$$

which suggests positive skewness.

For a risk manager, negative skewness can therefore serve as an early warning signal for potentially severe downside events that may not be adequately represented by standard deviation alone.

10.2.5 Step-by-Step Example: Empirical Rule

Suppose the Bitsom Equity Fund has:

$$\mu = 1.88\%, \quad \sigma = 2.33\%.$$

Under the empirical rule, approximately 95% of observations should lie within:

$$\boxed{\mu \pm 2\sigma}$$

Substituting the values:

$$\begin{aligned} \mu \pm 2\sigma &= 1.88\% \pm 2(2.33\%) \\ &= 1.88\% \pm 4.66\%. \end{aligned}$$

Therefore:

$$\boxed{-2.78\% \leq R \leq 6.54\%}$$

Returns outside this interval would be considered statistically unusual under the normal-distribution assumption.

10.3 Probability and Association: The Mechanics of Diversification

Probability and association measures help analysts determine whether assets provide genuine diversification benefits or simply move together.

10.3.1 Pearson Correlation

Pearson correlation measures the strength and direction of a linear relationship between two variables.

$$\boxed{-1 \leq \rho_{XY} \leq 1}$$

The major interpretations are:

- $\rho = +1$: perfect positive co-movement,
- $\rho = 0$: no linear association,
- $\rho = -1$: perfect negative co-movement.

In portfolio construction, lower or negative correlations can provide greater diversification benefits.

10.3.2 Strategic Interpretation: Equity Cluster Risk

Suppose an eight-asset portfolio contains Nifty-50, Nifty IT, Nifty Bank, and Realty, with correlations between approximately 0.72 and 0.88.

Such high correlations suggest that these assets may behave similarly during market stress. Holding all four may therefore provide less diversification than their individual labels suggest.

If Nifty FMCG has correlations around 0.55–0.65 with the equity cluster, it may provide partial diversification.

Gold and USD/INR, when negatively correlated with the equity cluster, may provide stronger diversification benefits.

10.3.3 Probability Complement

The probability of an event not occurring is:

$$\boxed{P(A^c) = 1 - P(A)}$$

10.3.4 Union or Addition Rule

For two events A and B :

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The intersection is subtracted because it is counted twice when $P(A)$ and $P(B)$ are added.

10.3.5 Conditional Probability

Conditional probability measures the probability of event A occurring given that event B has already occurred.

$$\boxed{P(A | B) = \frac{P(A \cap B)}{P(B)}}$$

10.3.6 Independence

Two events are independent when the occurrence of one does not change the probability of the other.

$$P(A \cap B) = P(A)P(B)$$

10.3.7 Mutual Exclusivity

Two events are mutually exclusive if they cannot occur simultaneously.

$$P(A \cap B) = 0$$

10.3.8 Step-by-Step Example: Directionality in Cross-Tabulation

Consider a dataset containing 100 movies.

Suppose:

- 13 movies had blockbuster openings,
- 12 of those movies became hits,
- 30 movies were total hits.

The probability of a hit given a blockbuster opening is:

$$\begin{aligned} P(\text{Hit} \mid \text{Blockbuster Opening}) &= \frac{12}{13} \\ &= \boxed{92.3\%} \end{aligned}$$

However, the reverse conditional probability is:

$$\begin{aligned} P(\text{Blockbuster Opening} \mid \text{Hit}) &= \frac{12}{30} \\ &= \boxed{40.0\%} \end{aligned}$$

The two probabilities are not the same.

Strategic Insight: A blockbuster opening is a strong indicator of a hit because 92.3% of blockbuster openings result in hits. However, only 40% of all hits had blockbuster openings. Therefore, confusing $P(A \mid B)$ with $P(B \mid A)$ can lead to incorrect investment or business decisions.

10.4 Regression Modeling: Quantifying Predictive Power

Regression analysis is used to quantify relationships between a dependent variable and one or more explanatory variables.

10.4.1 Multiple Linear Regression

The general multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

where:

- Y = dependent variable,
- $X_1, \dots, X_n$ = explanatory variables,
- β_0 = intercept,
- β_i = coefficient of predictor X_i ,
- ϵ = error term.

Strategic Use: Regression can be used to determine how factors such as market returns, interest rates, or currency movements influence fund returns.

10.4.2 Ordinary Least Squares

Ordinary Least Squares (OLS) estimates the regression coefficients by minimizing the sum of squared residuals:

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value.

10.4.3 Coefficient of Determination

The coefficient of determination, R^2 , measures the proportion of variation in the dependent variable explained by the regression model.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

A higher R^2 generally indicates greater explanatory power, although it does not necessarily imply that the model is economically or statistically appropriate.

10.4.4 Adjusted R^2

Adjusted R^2 penalizes the model for adding unnecessary explanatory variables.

$$\bar{R}^2 = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

where:

- n = number of observations,
- k = number of explanatory variables.

Strategic Use: Adjusted R^2 is particularly useful in multiple regression because ordinary R^2 can increase simply by adding more predictors.

10.4.5 Logistic Regression

Logistic regression is appropriate when the dependent variable is binary, such as:

$$Y = \begin{cases} 1, & \text{Default,} \\ 0, & \text{Paid.} \end{cases}$$

The logistic model is:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n$$

The output is a probability between 0 and 1.

Strategic Use: Logistic regression is widely used for credit scoring, fraud detection, customer churn, and other binary classification problems.

10.4.6 Strategic Interpretation: Fit versus Complexity

Suppose a regression model explains approximately 69% of the variation in fund returns. Although this may appear strong, adding irrelevant predictors can artificially increase R^2 .

Therefore, adjusted R^2 should be examined when multiple predictors are used.

For binary outcomes such as “Loan Default” versus “Loan Paid”, linear regression is generally inappropriate because the predicted output is not naturally constrained between 0 and 1. Logistic regression solves this problem by producing a probability.

10.4.7 Step-by-Step Example: Predicting Fund Return

Consider the model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3$$

with:

$$\beta_0 = 0.42,$$

$$X_1 = 1.8\%, \quad \beta_1 = 0.82,$$

and

$$X_3 = 0.35\%, \quad \beta_3 = -0.35.$$

Substituting the values:

$$\begin{aligned} Y &= 0.42 + (0.82)(1.8) - (0.35)(0.35) \\ &= 0.42 + 1.476 - 0.1225 \\ &= \boxed{1.7735\%}. \end{aligned}$$

Thus, the model predicts a return of approximately:

$$1.77\%$$

which may be reported as approximately 1.8% after rounding.

10.5 Clustering and Distance Measures: Unsupervised Segmentation

Clustering is an unsupervised learning technique that groups observations according to their similarity. It is useful when predefined categories do not adequately capture the underlying structure of the data.

10.5.1 Euclidean Distance

Euclidean distance represents the straight-line distance between two points.

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Strategic Use: It is widely used in K-means clustering. However, it is sensitive to differences in variable scale.

10.5.2 Manhattan Distance

Manhattan distance calculates the sum of absolute differences.

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|$$

It is sometimes called city-block distance.

Strategic Use: Manhattan distance can be useful when absolute differences are more meaningful or when robustness to individual large coordinate differences is desired.

10.5.3 Jaccard Similarity

The Jaccard similarity coefficient is commonly used for binary or categorical data.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

The value ranges between 0 and 1:

$$0 \leq J(A, B) \leq 1.$$

A value of 1 indicates complete similarity, while 0 indicates no shared elements.

10.5.4 Calinski–Harabasz Index

The Calinski–Harabasz (CH) index evaluates the quality of a clustering solution by comparing between-cluster dispersion with within-cluster dispersion.

A general representation is:

$$CH = \frac{\text{Tr}(B_k)/(k-1)}{\text{Tr}(W_k)/(n-k)}$$

where:

- B_k = between-cluster dispersion matrix,
- W_k = within-cluster dispersion matrix,
- k = number of clusters,
- n = number of observations.

A higher CH index generally indicates better-defined clusters.

10.5.5 Strategic Interpretation: Feature Scaling

Distance-based algorithms are sensitive to the scale of input variables.

For example, in a wine dataset, Alcohol may be measured in single digits, while Alkalinity of Ash may have values in the hundreds. Without standardization, the latter variable can dominate Euclidean distance.

A common standardization method is the Z-score:

$$z_i = \frac{x_i - \mu}{\sigma}$$

Standardization places variables on a comparable scale and prevents a large-scale variable from dominating the clustering algorithm.

10.5.6 Step-by-Step Example: K-Means Assignment

Suppose a new observation has the following distances from three centroids:

$$d(C_1) = 46, \quad d(C_2) = 10, \quad d(C_3) = 9.$$

The nearest centroid is C_3 because:

$$9 < 10 < 46.$$

Therefore:

$$\text{Assignment} = C_3$$

The objective of K-means is to minimize the within-cluster sum of squared Euclidean distances.

10.6 Model Evaluation Metrics: Validating Analytical Rigor

Model evaluation determines whether a model performs well on unseen data. A model that performs extremely well on training data but poorly on new data may be suffering from overfitting.

10.6.1 Precision

Precision measures the proportion of predicted positive observations that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

where:

- TP = True Positives,
- FP = False Positives.

Interpretation: Precision answers the question:

“Of all observations predicted to be positive, how many were actually positive?”

10.6.2 Recall

Recall measures the proportion of actual positive observations that the model successfully identifies.

$$\text{Recall} = \frac{TP}{TP + FN}$$

where FN represents False Negatives.

Interpretation: Recall answers the question:

“Of all actual positive cases, how many did the model successfully detect?”

10.6.3 F1 Score

The F1 score is the harmonic mean of precision and recall.

$$F_1 = 2 \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

It is particularly useful when both false positives and false negatives are important.

10.6.4 Mean Absolute Error

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where:

- y_i = actual value,
- $\hat{y}_i$ = predicted value.

Interpretation: MAE is easy to interpret because it is expressed in the same units as the dependent variable.

10.6.5 Root Mean Squared Error

RMSE measures the square root of the average squared prediction error.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Because errors are squared, RMSE gives greater weight to large prediction errors than MAE.

10.6.6 Strategic Interpretation: Diagnostic RMSE

Suppose a credit-risk model achieves:

$$\text{Training Accuracy} = 98\%$$

but only:

$$\text{Production Accuracy} = 61\%.$$

The large performance deterioration suggests substantial overfitting. The model may have learned patterns specific to the historical training dataset rather than generalizable relationships.

Similarly, if a return forecasting model has:

$$RMSE = 1.44\%,$$

the typical magnitude of the model's prediction error, measured through RMSE, is approximately 1.44 percentage points, or:

$$144 \text{ basis points}$$

This provides a useful benchmark for evaluating the economic usefulness of the forecast.

10.6.7 Step-by-Step Example: Fraud Detection

Suppose a fraud detection system produces the following results:

- Actual fraudulent transactions = 10,
- Predicted fraudulent transactions = 8,
- $TP = 8$,
- $FN = 2$,
- Actual legitimate transactions = 90,
- $TN = 90$,
- $FP = 2$.

The precision is:

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP + FP} \\ &= \frac{8}{8 + 2} \\ &= \boxed{80\%}\end{aligned}$$

The recall is:

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{8}{8 + 2} \\ &= \boxed{80\%}\end{aligned}$$

The F1 score is therefore:

$$\begin{aligned}F_1 &= 2 \left(\frac{0.80 \times 0.80}{0.80 + 0.80} \right) \\ &= \boxed{0.80}\end{aligned}$$

or:

$$\boxed{F_1 = 80\%}.$$

10.7 Final Strategic Principle: “Visualize First”

Regardless of the sophistication of a mathematical model, statistical analysis should be supported by appropriate visualization.

For example:

- correlation heat maps can reveal clusters of highly correlated assets;
- histograms and density plots can reveal skewness and fat tails;

- box plots can identify extreme observations;
- scatter plots can reveal nonlinear relationships;
- residual plots can identify violations of regression assumptions;
- cluster plots can help evaluate the quality of segmentation.

The central principle is therefore:

Visualize $\longrightarrow$ Diagnose $\longrightarrow$ Model $\longrightarrow$ Validate $\longrightarrow$ Interpret

Mathematical models should not be treated as isolated formulas. Their usefulness depends on whether their underlying assumptions are consistent with the observed data and the economic problem being analyzed.

11 Capital Markets and Asset Pricing Formulas

This section summarizes the key mathematical models used in capital markets, asset pricing, derivatives, fintech credit risk, corporate valuation, and agentic AI-based credit scoring. Each formula is followed by a brief explanation and a numerical example to demonstrate its practical application.

11.1 Systematic Risk: Beta

Formula:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

Explanation:

Beta measures the systematic, non-diversifiable risk of an individual asset relative to the overall market. It indicates how sensitive the asset's return is to movements in the market.

- $\beta = 1$: The asset moves approximately in line with the market.
- $\beta > 1$: The asset is more sensitive to market movements.
- $\beta < 1$: The asset is less sensitive to market movements.
- $\beta < 0$: The asset tends to move in the opposite direction to the market.

Use Case:

Beta is useful for measuring exposure to systematic risks such as macroeconomic shocks, interest-rate changes, recessions, geopolitical events, and changes in market sentiment.

Example:

Utility companies typically have a $\beta < 1$ because their revenues tend to be relatively stable across business cycles. Conversely, a highly cyclical technology or software-export company may have a $\beta > 1$.

11.2 Capital Asset Pricing Model (CAPM)

Formula:

$$E[R_i] = R_f + \beta_i (R_m - R_f)$$

where:

- $E[R_i]$ = expected return of asset i ,
- R_f = risk-free rate,
- β_i = beta of asset i ,
- R_m = expected market return,
- $R_m - R_f$ = market risk premium.

Explanation:

CAPM estimates the return investors should require from an asset based on its systematic risk. The required return consists of the risk-free rate plus a risk premium for bearing market risk.

Example:

Suppose:

$$R_f = 5\%, \quad R_m = 13\%.$$

Therefore, the market risk premium is:

$$R_m - R_f = 13\% - 5\% = 8\%.$$

For $\beta = 1$:

$$E[R] = 5\% + 1(8\%) = \boxed{13\%}.$$

For $\beta = 2$:

$$E[R] = 5\% + 2(8\%) = \boxed{21\%}.$$

Thus, the higher-beta stock requires a higher expected return as compensation for taking greater systematic risk.

11.3 Actual Stock Return

Formula:

$$R_i = \frac{P_{t+1} - P_t}{P_t}$$

If dividends are included, the total holding-period return can be written as:

$$R_i = \frac{P_{t+1} - P_t + D}{P_t}$$

where D represents dividends received during the period.

Explanation:

The formula measures the percentage gain or loss generated by an investment. It captures the change in the market price and, when applicable, income received through dividends.

Example:

Suppose an investor buys a stock at:

$$P_t = 2,950$$

and sells it at:

$$P_{t+1} = 3,097.$$

Then:

$$R_i = \frac{3097 - 2950}{2950} = \frac{147}{2950} = 0.05.$$

Therefore:

$$\boxed{R_i = 5\%}$$

11.4 Alpha: Mispricing Signal

Formula:

$$\boxed{\alpha_i = R_i - E[R_i]_{\text{CAPM}}}$$

Explanation:

Alpha measures the difference between an asset's actual return and the return predicted by CAPM for its level of systematic risk.

- $\alpha > 0$: Actual return exceeds the CAPM-required return.
- $\alpha = 0$: Actual return equals the CAPM-required return.
- $\alpha < 0$: Actual return is below the CAPM-required return.

Example:

Suppose:

$$R_m = 13\%, \quad R_f = 7\%.$$

For Portfolio A:

$$\beta = 1.$$

Its CAPM expected return is:

$$E[R_A] = 7\% + 1(13\% - 7\%) = 13\%.$$

If the actual return is 12%:

$$\alpha_A = 12\% - 13\% = -1\%.$$

Hence:

$$\boxed{\alpha_A = -1\%}$$

Similarly, if Portfolio C has:

$$\beta = 0.367$$

then:

$$E[R_C] = 7\% + 0.367(13\% - 7\%) \approx 9.2\%.$$

If its actual return is 10.2%:

$$\alpha_C = 10.2\% - 9.2\% = \boxed{+1.0\%}.$$

A positive alpha indicates outperformance relative to the CAPM benchmark.

11.5 Bid–Ask Spread

Formula:

$$\boxed{\text{Spread} = \text{Best Ask Price} - \text{Best Bid Price}}$$

The mid-price is:

$$\boxed{\text{Mid-Price} = \frac{\text{Best Ask} + \text{Best Bid}}{2}}$$

Explanation:

The bid–ask spread measures the difference between the price at which a trader can immediately sell and the price at which the trader can immediately buy. A smaller spread generally indicates greater liquidity.

Example:

Suppose:

$$\text{Best Bid} = 210.50$$

and

$$\text{Best Ask} = 211.75.$$

Then:

$$\text{Spread} = 211.75 - 210.50 = \boxed{1.25}.$$

The mid-price is:

$$\text{Mid-Price} = \frac{211.75 + 210.50}{2} = \boxed{211.125}.$$

11.6 Option Payoffs at Expiry

11.6.1 Call Option

Formula:

$$\boxed{\text{Call Profit} = \max(S_T - K, 0) - C}$$

11.6.2 Put Option

Formula:

$$\text{Put Profit} = \max(K - S_T, 0) - C$$

where:

- S_T = stock price at expiry,
- K = strike price,
- C = option premium paid.

Explanation:

A call option gives the buyer the right to buy the underlying asset at the strike price. A put option gives the buyer the right to sell the underlying asset at the strike price.

The premium paid for the option must be deducted when calculating the buyer's net profit.

Example: Call Option

Suppose:

$$K = 100, \quad C = 1.$$

If:

$$S_T = 105,$$

then:

$$\text{Call Profit} = \max(105 - 100, 0) - 1 = 5 - 1 = \boxed{4}.$$

If:

$$S_T = 99,$$

then:

$$\text{Call Profit} = \max(99 - 100, 0) - 1 = 0 - 1 = \boxed{-1}.$$

Thus, when the option expires out-of-the-money, the buyer loses the premium.

11.7 Put–Call Parity

Formula:

For European options without dividends:

$$C - P = S_0 - PV(K)$$

Equivalently:

$$C + PV(K) = P + S_0$$

where:

- C = European call price,
- P = European put price,

- S_0 = current stock price,
- $PV(K)$ = present value of the strike price.

Explanation:

Put–call parity is a no-arbitrage relationship. Two portfolios with identical future cash flows must have identical prices today.

Consider a stock with:

$$S_0 = 70, \quad K = 70.$$

A protective put consists of:

$$\text{Stock} + \text{Put}.$$

At expiry, if:

$$S_T = 40,$$

the stock is worth 40 and the put is worth:

$$70 - 40 = 30.$$

Therefore:

$$40 + 30 = \boxed{70}.$$

If:

$$S_T = 100,$$

the put expires worthless and:

$$100 + 0 = \boxed{100}.$$

This payoff can be replicated by:

$$\text{Call} + \text{Risk-Free Bond}.$$

Therefore, put–call parity prevents arbitrage opportunities between these equivalent portfolios.

12 Fintech Credit Risk and Fraud Detection Formulas

12.1 Linear Regression Risk Score

Formula:

$$\boxed{\text{Score} = b_0 + w_1X_1 + w_2X_2 + \cdots + w_nX_n}$$

Explanation:

A linear scoring model combines several borrower characteristics using different weights. Each variable contributes to the final risk score according to its coefficient.

For example:

$$w_{\text{income}} < 0$$

may indicate that higher income reduces credit risk, while:

$$w_{\text{bounces}} > 0$$

may indicate that more bounced payments increase credit risk.

Example:

A simplified model could be:

$$\text{Score} = b_0 - 0.002(\text{Income}) + 0.50(\text{Bounces}).$$

A borrower with higher income would therefore receive a lower risk score, while bounced payments would increase the score.

12.2 Logistic Regression and Probability of Default

Formula:

$$P(\text{Default}) = \frac{1}{1 + e^{-\text{Score}}}$$

More generally:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

Explanation:

Logistic regression converts a linear score into a probability between 0 and 1. This makes it particularly useful for binary credit outcomes such as:

$$Y = \begin{cases} 1, & \text{Default,} \\ 0, & \text{No Default.} \end{cases}$$

If the score is highly negative:

$$P(\text{Default}) \rightarrow 0.$$

If the score is highly positive:

$$P(\text{Default}) \rightarrow 1.$$

12.3 Differential Risk-Adjusted Pricing

Formula:

$$\text{Interest Rate} = \text{Base Rate} + (\text{Spread} \times P(\text{Default}))$$

Explanation:

Risk-based pricing allows a lender to charge different interest rates depending on the estimated probability of default. A higher probability of default results in a higher risk premium.

Example:

Suppose:

$$\text{Base Rate} = 12\%, \quad \text{Spread} = 24\%.$$

For:

$$P(\text{Default}) = 0.5,$$

the interest rate is:

$$12\% + 24\%(0.5) = 12\% + 12\% = \boxed{24\%}.$$

For:

$$P(\text{Default}) = 1,$$

the rate becomes:

$$12\% + 24\%(1) = \boxed{36\%}.$$

13 Corporate Valuation Formulas

13.1 Free Cash Flow to the Firm (FCFF)

Formula:

$$FCF = EBIT(1 - t) + \text{Depreciation} - CapEx - \Delta NWC$$

where:

- $EBIT$ = Earnings Before Interest and Taxes,
- t = corporate tax rate,
- $CapEx$ = capital expenditure,
- ΔNWC = change in net working capital.

An equivalent formulation using operating cash flow is:

$$FCF = \text{Operating Cash Flow} - CapEx - \Delta NWC$$

Explanation:

Free cash flow represents the operating cash generated by a business after paying taxes and making the investments necessary to maintain and grow its operations. It is available to both debt and equity providers.

Example:

Suppose:

$$EBIT = \$219 \text{ million},$$

$$\text{Depreciation} = \$90 \text{ million},$$

$$\text{Taxes} = \$71 \text{ million},$$

$$\text{CapEx} = \$173 \text{ million},$$

and:

$$\Delta NWC = \$23 \text{ million}.$$

Operating cash flow is:

$$OCF = 219 + 90 - 71 = \$238 \text{ million}.$$

Therefore:

$$FCF = 238 - 173 - 23 = \boxed{\$42 \text{ million}}.$$

13.2 After-Tax Cost of Debt

Formula:

$$\boxed{R_d = R_{\text{interest}}(1 - t)}$$

Explanation:

Interest expenses generally provide a tax shield because they are deductible for tax purposes. Therefore, the effective cost of debt after tax is lower than the stated interest rate.

Example:

Suppose:

$$R_{\text{interest}} = 8.5\%$$

and:

$$t = 25\%.$$

Then:

$$R_d = 8.5\%(1 - 0.25)$$

$$= 8.5\% \times 0.75$$

$$= \boxed{6.375\% \approx 6.4\%}.$$

13.3 Weighted Average Cost of Capital (WACC)

Formula:

$$WACC = \left(\frac{E}{V}\right) R_e + \left(\frac{D}{V}\right) R_d$$

where:

$$V = E + D$$

and:

- E = market value of equity,
- D = market value of debt,
- V = total market value of capital,
- R_e = cost of equity,
- R_d = after-tax cost of debt.

Explanation:

WACC represents the average rate of return required by all providers of capital. It is commonly used as the discount rate when valuing a firm's future free cash flows.

Example:

Suppose:

$$R_e = 13.5\%, \quad R_d = 6.4\%,$$

with:

$$\frac{E}{V} = 75\%, \quad \frac{D}{V} = 25\%.$$

Then:

$$\begin{aligned} WACC &= (0.75)(13.5\%) + (0.25)(6.4\%) \\ &= 10.125\% + 1.60\% \\ &= \boxed{11.725\% \approx 11.73\%}. \end{aligned}$$

13.4 Constant-Growth Perpetuity and Terminal Value

Formula:

$$TV_5 = \frac{FCF_5(1+g)}{WACC - g}$$

where:

- TV_5 = terminal value at the end of Year 5,
- FCF_5 = free cash flow in Year 5,

- g = perpetual growth rate,
- $WACC$ = weighted average cost of capital.

The condition:

$$\boxed{WACC > g}$$

must hold for the formula to produce a meaningful finite value.

Explanation:

Terminal value represents the value of all cash flows generated after the explicit forecast period. The constant-growth model assumes that cash flows grow at a stable rate forever after the final forecast year.

Example:

Suppose:

$$FCF_5 = 682 \text{ crore,}$$

$$WACC = 11.74\%, \quad g = 5\%.$$

Then:

$$TV_5 = \frac{682(1.05)}{0.1174 - 0.05}.$$

Therefore:

$$TV_5 = \frac{716.1}{0.0674} \approx \boxed{10,625 \text{ crore}}.$$

14 Agentic AI Custom Scoring Logic

14.1 Mock Loan Officer Risk Score

Algorithm:

$$\begin{aligned} \text{Raw Score} = & 20 + 25 \mathbf{1}(\text{CIBIL} > 700) \\ & + 12 \mathbf{1}(\text{Salary Credited}) \\ & - 10 \mathbf{1}(\text{Bounced Payments}) \\ & + 13 \mathbf{1}(\text{Monthly Surplus} > 30,000) \end{aligned}$$

The final score is constrained to the interval $[0, 100]$:

$$\boxed{\text{Score} = \min(100, \max(0, \text{Raw Score}))}$$

where $\mathbf{1}(\cdot)$ is an indicator function that takes the value 1 when the condition is satisfied and 0 otherwise.

Risk Classification:

$$\begin{cases} \text{Low Risk,} & \text{Score} > 85, \\ \text{Medium Risk,} & 65 \leq \text{Score} \leq 85, \\ \text{High Risk,} & \text{Score} < 65. \end{cases}$$

Explanation:

This is a deterministic scoring rule that can be incorporated into an agentic AI lending workflow. It converts borrower characteristics into a standardized score before additional policy checks, human review, or loan approval decisions.

Example 1:

Suppose an applicant has:

$$\text{CIBIL} = 742,$$

confirmed salary credit, zero bounced payments, and monthly surplus of:

$$76,350.$$

Therefore:

$$\begin{aligned} \text{Score} &= 20 + 25 + 12 - 0 + 13 \\ &= \boxed{70}. \end{aligned}$$

Hence:

$$\boxed{\text{Medium Risk}}$$

because:

$$65 \leq 70 \leq 85.$$

Example 2:

Suppose another applicant has:

$$\text{CIBIL} = 720,$$

salary credited, one bounced payment, and monthly surplus of:

$$50,000.$$

Then:

$$\begin{aligned} \text{Score} &= 20 + 25 + 12 - 10 + 13 \\ &= \boxed{60}. \end{aligned}$$

Therefore:

$$\boxed{\text{High Risk}}$$

because:

$$60 < 65.$$

15 Quick Formula Summary

For rapid revision, the most important formulas are:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

$$E[R_i] = R_f + \beta_i(R_m - R_f)$$

$$\alpha_i = R_i - E[R_i]_{\text{CAPM}}$$

$$R_i = \frac{P_{t+1} - P_t}{P_t}$$

$$\text{Spread} = \text{Ask} - \text{Bid}$$

$$\text{Call Profit} = \max(S_T - K, 0) - C$$

$$\text{Put Profit} = \max(K - S_T, 0) - C$$

$$C - P = S_0 - PV(K)$$

$$P(\text{Default}) = \frac{1}{1 + e^{-\text{Score}}}$$

$$\text{Interest Rate} = \text{Base Rate} + \text{Spread} \times P(\text{Default})$$

$$FCF = EBIT(1 - t) + \text{Depreciation} - CapEx - \Delta NWC$$

$$R_d = R_{\text{interest}}(1 - t)$$

$$WACC = \frac{E}{V}R_e + \frac{D}{V}R_d$$

$$TV_5 = \frac{FCF_5(1 + g)}{WACC - g}$$

$$\text{Score} = \min(100, \max(0, \text{Raw Score}))$$

16 Summary of Financial and Cryptocurrency Analytics Formulas

This section summarizes the key mathematical formulas used in cryptocurrency analysis, financial economics, asset pricing, and decentralized finance (DeFi). For each formula, the mathematical expression is followed by a brief explanation of its interpretation, use case, and a worked example.

16.1 Daily Return of a Cryptocurrency

The daily return measures the percentage change in the price of a cryptocurrency from one day to the next.

$$R_t = \frac{P_{t+1} - P_t}{P_t} \quad (1)$$

An equivalent representation is:

$$R_t = \frac{P_{t+1}}{P_t} - 1 \quad (2)$$

where:

- P_t is the cryptocurrency price at time t ,
- P_{t+1} is the cryptocurrency price at time $t + 1$,
- R_t is the daily return.

Explanation: A positive value of R_t represents a price increase, while a negative value represents a price decline. Daily returns are commonly used as the basic input for volatility, correlation, beta, Sharpe ratio, and other financial risk measures.

Example:

Suppose Bitcoin increases by 3% from one day to the next. Then:

$$R_t = 0.03 = 3\% \quad (3)$$

If the average daily Bitcoin return over a sample period is approximately 0.25%, then a \$1 investment grows on average to approximately:

$$\$1(1 + 0.0025) = \$1.0025 \quad (4)$$

Thus, the average daily return represents the typical one-day percentage change during the observed period.

16.2 Purchasing Power Loss Due to Inflation

Inflation reduces the quantity of goods and services that can be purchased with a fixed amount of money. The purchasing power retained can be expressed as:

$$\text{Purchasing Power Retained} = \frac{\text{Original Price}}{\text{New Price}} \quad (5)$$

Therefore, purchasing power loss is:

$$\text{Purchasing Power Loss} = 1 - \frac{\text{Original Price}}{\text{New Price}} \quad (6)$$

Explanation: This formula measures how much of the original purchasing power remains after prices increase.

Example:

Suppose an item costs 100 today but costs 120 in the future. The purchasing power retained is:

$$\frac{100}{120} = 0.8333 = 83.33\% \quad (7)$$

Therefore, purchasing power loss is:

$$1 - 0.8333 = 0.1667 = 16.67\% \quad (8)$$

Thus, although the nominal amount is still 100, its purchasing power has declined by approximately 16.67% relative to the new price level.

16.3 Sharpe Ratio

The Sharpe ratio measures risk-adjusted performance by comparing the excess return of an investment with its total volatility.

$$\text{Sharpe Ratio} = \frac{E[R] - R_f}{\sigma_R} \quad (9)$$

where:

- $E[R]$ is the expected or average return,
- R_f is the risk-free rate,
- σ_R is the standard deviation of returns.

When the risk-free rate is assumed to be zero:

$$\text{Sharpe Ratio} = \frac{E[R]}{\sigma_R} \quad (10)$$

Explanation: The Sharpe ratio indicates how much return an investor receives for each unit of total risk. A higher Sharpe ratio generally indicates better risk-adjusted performance.

Example:

Suppose Bitcoin has an average daily return of 0.253% and a standard deviation of 4.37%. Assuming $R_f = 0$:

$$\text{Sharpe}_{BTC} = \frac{0.253\%}{4.37\%} \approx 0.058 \quad (11)$$

For Ether, suppose the average daily return is 0.579% and the standard deviation is 7.22%:

$$\text{Sharpe}_{ETH} = \frac{0.579\%}{7.22\%} \approx 0.080 \quad (12)$$

Therefore, Ether has the higher daily Sharpe ratio in this example, indicating greater return per unit of total volatility.

16.4 Capital Asset Pricing Model (CAPM)

The Capital Asset Pricing Model estimates the expected return of an asset based on its systematic risk relative to the market.

$$E[R_i] = R_f + \beta_i (E[R_M] - R_f) \quad (13)$$

where:

- $E[R_i]$ is the expected return of asset i ,
- R_f is the risk-free rate,
- β_i is the systematic risk of asset i ,
- $E[R_M]$ is the expected market return,
- $E[R_M] - R_f$ is the market risk premium.

Explanation: CAPM separates expected return into two components: compensation for the time value of money, represented by R_f , and compensation for systematic market risk, represented by $\beta_i(E[R_M] - R_f)$.

Example:

Suppose:

$$R_f = 5\%, \quad E[R_M] = 13\% \quad (14)$$

Therefore, the market risk premium is:

$$E[R_M] - R_f = 13\% - 5\% = 8\% \quad (15)$$

For an asset with $\beta = 1$:

$$E[R_i] = 5\% + 1(8\%) = 13\% \quad (16)$$

For an asset with $\beta = 2$:

$$E[R_i] = 5\% + 2(8\%) = 21\% \quad (17)$$

A higher beta therefore implies a higher required expected return under the CAPM framework.

16.5 Beta (β)

Beta measures the systematic sensitivity of an asset's return to movements in the market.

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)} \quad (18)$$

where:

- $\text{Cov}(R_i, R_M)$ is the covariance between the asset and market returns,
- $\text{Var}(R_M)$ is the variance of market returns.

Explanation: Beta indicates how strongly an asset tends to move with the market.

- $\beta = 1$: The asset tends to move approximately in line with the market.
- $\beta > 1$: The asset tends to be more sensitive to market movements.
- $\beta < 1$: The asset tends to be less sensitive to market movements.
- $\beta < 0$: The asset tends to move inversely to the market.

Thus, beta is primarily a measure of *systematic* rather than diversifiable risk.

16.6 Alpha (α) or Abnormal Return

Alpha measures the difference between an asset's observed return and the return predicted by the CAPM.

$$\alpha = E[R_i] - [R_f + \beta_i (E[R_M] - R_f)] \quad (19)$$

Explanation: A positive alpha indicates that the observed return is higher than the CAPM-required return, whereas a negative alpha indicates underperformance relative to the CAPM benchmark.

Example:

Suppose the market return is 13% and the risk-free rate is 7%.

For Portfolio A:

$$\beta_A = 1.0 \quad (20)$$

The CAPM expected return is:

$$E[R_A] = 7\% + 1(13\% - 7\%) = 13\% \quad (21)$$

If its actual return is 12%, then:

$$\alpha_A = 12\% - 13\% = -1\% \quad (22)$$

Therefore, Portfolio A underperformed its CAPM benchmark by 1 percentage point. For Portfolio C, suppose:

$$\beta_C = 0.367 \quad (23)$$

Then:

$$E[R_C] = 7\% + 0.367(6\%) \approx 9.2\% \quad (24)$$

If the actual return is 10.2%:

$$\alpha_C = 10.2\% - 9.2\% = +1.0\% \quad (25)$$

This indicates positive abnormal performance relative to CAPM.

16.7 Gordon Growth Model

The Gordon Growth Model estimates the intrinsic value of an asset based on dividends growing at a constant rate.

$$P_0 = \frac{D_1}{r - g} \quad (26)$$

where:

- P_0 is the current intrinsic value,
- D_1 is the expected dividend in the next period,
- r is the required rate of return,
- g is the constant dividend growth rate.

Explanation: The model is particularly appropriate for mature companies whose dividends are expected to grow at a relatively stable rate.

The condition

$$r > g \quad (27)$$

must hold for the model to produce a finite positive value.

For an asset that produces no dividends, such as a cryptocurrency under a strict dividend-based interpretation:

$$D_1 = 0 \implies P_0 = 0 \quad (28)$$

This illustrates a limitation of applying traditional dividend-discount models directly to non-dividend-paying assets. Their economic value, if any, must be explained through other valuation mechanisms.

16.8 Collateralization Ratio

The collateralization ratio measures the amount of collateral supporting a loan or stablecoin issuance.

$$\text{Collateralization Ratio} = \frac{\text{Value of Collateral}}{\text{Value of Loans or Stablecoins Issued}} \quad (29)$$

Explanation: A ratio above 100% indicates overcollateralization. The larger the ratio, the larger the collateral buffer available to absorb declines in the collateral's market value.

Example:

Suppose a DeFi protocol issues \$150,000 of stablecoins and holds \$300,000 of cryptocurrency collateral:

$$\text{Collateralization Ratio} = \frac{300,000}{150,000} = 2 = 200\% \quad (30)$$

Thus, the position is collateralized at 200%.

Interpretation: For every \$1 of stablecoin issued, the protocol holds \$2 of collateral. This provides a substantial buffer against collateral-price declines, although it does not eliminate liquidation or smart-contract risks.

16.9 Summary Interpretation

The formulas above connect several major concepts in financial analytics:

1. **Daily return** measures short-term price performance.
2. **Purchasing power loss** measures the effect of inflation on the real value of money.
3. **Sharpe ratio** evaluates return relative to total risk.
4. **CAPM** estimates the required return associated with systematic market risk.
5. **Beta** measures sensitivity to market movements.
6. **Alpha** measures performance relative to the CAPM benchmark.
7. **Gordon Growth Model** values assets based on growing dividend streams.
8. **Collateralization ratio** measures the financial buffer supporting loans or stable-coin issuance.

Together, these measures provide a framework for evaluating return, risk, valuation, and financial stability across traditional and cryptocurrency markets.

17 Levels of Measurement: Nominal, Ordinal, Interval, and Ratio Data

Understanding the level of measurement is one of the first steps in statistical analysis. The four major levels of measurement are **nominal**, **ordinal**, **interval**, and **ratio**. They describe the mathematical properties of the data and determine which statistical operations are appropriate.

17.1 What is Data?

Data refers to information collected for the purpose of analysis. For example, if information is collected from five students, the dataset might contain variables such as gender, class rank, temperature, and weight.

- Gender: Male, Female, Female, Male, Female
- Class rank: 1st, 2nd, 3rd, 1st, 2nd
- Temperature: 20°C , 25°C , 30°C , 22°C , 28°C
- Weight: 50 kg, 60 kg, 70 kg, 55 kg, 80 kg

Although all of these are data, they do not have the same mathematical properties. Therefore, data are classified into different levels of measurement.

17.2 The Four Levels of Measurement

The four levels can be viewed as a progression:

$$\boxed{\text{Nominal} \rightarrow \text{Ordinal} \rightarrow \text{Interval} \rightarrow \text{Ratio}}$$

As we move from nominal to ratio, the number of mathematical operations that can be meaningfully performed on the data increases.

17.2.1 Nominal Data

Nominal data consists of names, labels, or categories with no inherent ordering.

Examples include:

- Blood group: A, B, AB, O
- Eye color: Brown, Blue, Green
- Gender
- Religion
- Type of car

For example, consider:

$$\text{Eye Color} = \{\text{Brown, Brown, Blue, Green, Blue}\}.$$

These observations can be counted, but they cannot be meaningfully ranked.

$$\text{Brown} > \text{Blue}$$

has no mathematical meaning.

Similarly, calculating the arithmetic mean of eye color is meaningless.

Key idea:

$$\boxed{\text{Nominal} = \text{Name or Category}}$$

What can be done?

- Frequency/count: ✓
- Ordering: ×
- Median: ×
- Mean: ×
- Standard deviation: ×

17.2.2 Ordinal Data

Ordinal data consists of categories that have a meaningful order or ranking, but the differences between adjacent categories are not necessarily equal.

Examples include:

- Satisfaction: Poor $\rightarrow$ Fair $\rightarrow$ Good $\rightarrow$ Excellent
- Pain: Mild $\rightarrow$ Moderate $\rightarrow$ Severe
- Class rank: 1st $\rightarrow$ 2nd $\rightarrow$ 3rd
- Education level: Primary $\rightarrow$ Secondary $\rightarrow$ College

For example, satisfaction may be coded as:

$$1 = \text{Poor}, \quad 2 = \text{Fair}, \quad 3 = \text{Good}, \quad 4 = \text{Excellent}.$$

The ordering is meaningful:

$$\text{Excellent} > \text{Good} > \text{Fair} > \text{Poor}.$$

However, we cannot assume that the difference between Poor and Fair is exactly the same as the difference between Good and Excellent.

Therefore:

- Ordering: ✓
- Median: ✓
- Percentiles: ✓
- Mean: Usually not meaningful
- Standard deviation: Usually not meaningful

Key idea:

$$\text{Ordinal} = \text{Category} + \text{Order}$$

17.2.3 Interval Data

Interval data consists of numerical observations where the differences between values are equal, but there is no meaningful absolute zero.

The classic example is temperature measured in degrees Celsius.

Consider:

$$10^{\circ}\text{C}, \quad 20^{\circ}\text{C}, \quad 30^{\circ}\text{C}.$$

The difference between 10°C and 20°C is:

$$20 - 10 = 10^{\circ}\text{C}.$$

Similarly:

$$30 - 20 = 10^{\circ}C.$$

Therefore, the intervals are equal.

However, $0^{\circ}C$ does not represent the complete absence of temperature. Consequently, statements such as

$$20^{\circ}C = 2(10^{\circ}C)$$

do not have a meaningful ratio interpretation.

Interval data therefore supports:

- Addition and subtraction: ✓
- Mean: ✓
- Standard deviation: ✓
- Meaningful ratios such as “twice as much”: ✗

Key idea:

Interval = Equal Differences, but No True Zero

17.2.4 Ratio Data

Ratio data has all the properties of interval data, but it also has a meaningful and absolute zero.

Examples include:

- Weight
- Height
- Age
- Distance
- Income
- Time duration

Consider weight:

$$0 \text{ kg} = \text{absence of weight.}$$

Therefore, ratio comparisons are meaningful:

$$20 \text{ kg} = 2(10 \text{ kg}).$$

Thus, ratio data permits essentially all standard arithmetic and statistical operations.

Key idea:

Ratio = Equal Differences + True Zero

Property	Nominal	Ordinal	Interval	Ratio
Frequency/count	✓	✓	✓	✓
Median/percentiles	×	✓	✓	✓
Add/subtract meaningfully	×	×	✓	✓
Mean/SD/SEM	×	Usually ×	✓	✓
Meaningful ratios/CV	×	×	×	✓

Table 1: Comparison of the four levels of measurement.

17.3 Summary of the Four Measurement Levels

17.4 Important Statistical Terms

Frequency refers to the number of times an observation or category occurs.

For example:

Number of male students = 10.

Median is the middle observation after arranging the data in ascending or descending order.

For example:

2, 4, 5, 7, 10

Therefore:

Median = 5.

Percentile indicates the relative position of an observation within a dataset. For example, being at the 90th percentile means that the observation is higher than approximately 90% of the observations.

Mean is the arithmetic average:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

Standard deviation measures the dispersion of observations around the mean:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}.$$

Standard Error of the Mean (SEM) measures the precision with which a sample mean estimates the population mean:

$$SEM = \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation and n is the sample size.

Coefficient of Variation (CV) measures standard deviation relative to the mean:

$$CV = \frac{\sigma}{\bar{x}} \times 100\%.$$

17.5 A Simple Example

Suppose information is collected about students:

- Name: Rahul, Priya, Amit
- Satisfaction: Poor, Good, Excellent
- Temperature: $10^{\circ}C$, $20^{\circ}C$, $30^{\circ}C$
- Weight: 20 kg, 40 kg, 60 kg

The corresponding measurement levels are:

Variable	Example	Measurement Level
Name	Rahul, Priya, Amit	Nominal
Satisfaction	Poor, Good, Excellent	Ordinal
Temperature	$10^{\circ}C$, $20^{\circ}C$, $30^{\circ}C$	Interval
Weight	20 kg, 40 kg, 60 kg	Ratio

Table 2: Simple examples of measurement levels.

17.6 Memory Trick: N–O–I–R

A simple way to remember the four levels is:

$$\boxed{N \rightarrow O \rightarrow I \rightarrow R}$$

- **N = Nominal:** Names
- **O = Ordinal:** Order
- **I = Interval:** Equal intervals
- **R = Ratio:** Real or true zero

Therefore:

Nominal	= Name
Ordinal	= Order
Interval	= Equal intervals
Ratio	= Equal intervals + true zero

17.7 Categorical and Quantitative Data

Data can also be broadly divided into two major groups:

$$\boxed{\text{Data} \rightarrow \begin{cases} \text{Categorical Data} \\ \text{Quantitative Data} \end{cases}}$$

17.7.1 Categorical Data

Categorical data places observations into groups or categories.

For example:

$$\text{Blood Group} = \{A, B, AB, O\}.$$

Categorical data may be represented using words or numbers.

For example:

$$1 = \text{Male}, \quad 2 = \text{Female}.$$

Although 1 and 2 are numbers, they are being used only as labels. It would therefore be incorrect to interpret:

$$2 > 1$$

as meaning that Female is “greater than” Male.

Categorical data can be:

$$\text{Categorical} \rightarrow \begin{cases} \text{Nominal} \\ \text{Ordinal} \end{cases}$$

17.7.2 Quantitative Data

Quantitative data represents numerical quantities or amounts.

Examples include:

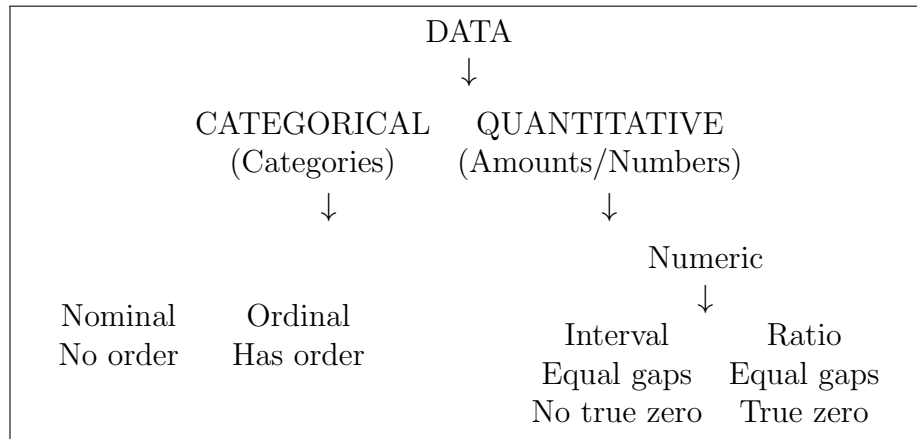
- Age = 25 years
- Weight = 60 kg
- Height = 170 cm
- Income = 50,000
- Temperature = $30^{\circ}C$

Quantitative data is generally divided into:

$$\text{Quantitative} \rightarrow \begin{cases} \text{Interval} \\ \text{Ratio} \end{cases}$$

17.8 Complete Classification Diagram

The complete classification can therefore be summarized as:



17.9 Important Correction: Numbers Do Not Always Mean Quantitative Data

A common source of confusion is assuming that numerical values automatically represent quantitative data.

This is not necessarily true.

For example:

1 = Poor, 2 = Fair, 3 = Good, 4 = Excellent.

These numbers represent ordered categories. They are therefore **ordinal**, not quantitative measurements.

The most useful question to ask is:

Do the numbers represent an actual amount, or are they simply labels?

This distinction prevents many common mistakes in statistical analysis.

18 Cross-Tabulation and Categorical Data Analysis

A **cross-tabulation**, or **crosstab**, is a table used to examine the relationship between two categorical variables.

In the movie example, the two variables are:

1. **Opening**: the amount earned by a movie during its opening period.
2. **Overall**: whether the movie eventually became a hit or a flop.

The main question is:

Is opening performance associated with overall movie performance?

Opening	Hit	Flop	Total
Blockbuster (> \$50 million)	12	1	13
Small (< \$50 million)	18	69	87
Total	30	70	100

Table 3: Cross-tabulation of opening performance and overall movie performance.

18.1 Movie Cross-Tabulation

Consider the following table:

There are 100 movies in total.

The columns represent the eventual outcome:

Hit = Movie earns more than \$100 million,

Flop = Movie earns less than \$100 million.

The rows represent opening performance:

Blockbuster opening > \$50 million,

Small opening < \$50 million.

18.2 Interpreting Individual Cells

Each cell represents the number of movies satisfying both conditions.

For example, the value 12 means:

12 movies had a blockbuster opening and eventually became hits.

The value 1 means:

1 movie had a blockbuster opening but eventually became a flop.

The value 18 means:

18 movies had a small opening but eventually became hits.

The value 69 means:

69 movies had a small opening and eventually became flops.

18.3 Checking the Row Totals

The first row contains:

$$12 + 1 = 13.$$

Therefore, there were 13 movies with blockbuster openings.

The second row contains:

$$18 + 69 = 87.$$

Therefore, there were 87 movies with small openings.
Hence:

$$13 + 87 = 100.$$

This confirms that all 100 movies are represented in the table.

18.4 Checking the Column Totals

The Hit column contains:

$$12 + 18 = 30.$$

Therefore, there were 30 hit movies.
The Flop column contains:

$$1 + 69 = 70.$$

Therefore, there were 70 flop movies.
Hence:

$$30 + 70 = 100.$$

The row and column totals are therefore internally consistent.

18.5 Understanding Crosstab Percentages

Percentages in a crosstab must always be interpreted according to their denominator.
Suppose the table reports:

$$12 \text{ (40\%)}.$$

The 40% is a **column percentage** because it is calculated using the total number of hit movies as the denominator:

$$\frac{12}{30} \times 100 = 40\%.$$

Therefore:

40% of all hit movies had blockbuster openings.

Similarly:

$$\frac{18}{30} \times 100 = 60\%.$$

Thus:

60% of all hit movies had small openings.

For the flop column:

$$\frac{1}{70} \times 100 \approx 1.43\%$$

and:

$$\frac{69}{70} \times 100 \approx 98.57\%.$$

Therefore, approximately:

1.5% of flop movies had blockbuster openings

and:

98.5% of flop movies had small openings.

18.6 Strategic Interpretation

The crosstab suggests that opening performance and eventual movie performance are associated.

In particular, almost all flop movies in this dataset had small openings:

98.5% of flop movies had small openings.

However, it is important not to confuse **association** with **causation**.

A crosstab demonstrates that two categorical variables occur together in a particular pattern, but by itself it does not prove that one variable causes the other.

Thus:

Crosstab $\Rightarrow$ Association, not necessarily Causation

18.7 Conditional Probability from the Crosstab

The same crosstab can be used to calculate conditional probabilities.

For example, the probability that a movie is a hit given that it had a blockbuster opening is:

$$P(\text{Hit} \mid \text{Blockbuster Opening}) = \frac{12}{13} \approx 92.3\%.$$

Therefore:

$$P(\text{Hit} \mid \text{Blockbuster Opening}) \approx 92.3\%$$

This means that among the 13 movies with blockbuster openings, approximately 92.3% eventually became hits.

The reverse conditional probability is different:

$$P(\text{Blockbuster Opening} \mid \text{Hit}) = \frac{12}{30} = 40\%.$$

Therefore:

$$P(\text{Blockbuster Opening} \mid \text{Hit}) = 40\%$$

These two probabilities are not the same.
This demonstrates an important principle:

$$P(A | B) \neq P(B | A)$$

in general.

18.8 How to Read Any Cross-Tabulation

Whenever a cross-tabulation is presented, follow these four steps:

1. **Identify the rows:** Determine the first categorical variable.
2. **Identify the columns:** Determine the second categorical variable.
3. **Read the cell counts:** Determine what each individual number represents.
4. **Check the percentages:** Identify the denominator used to calculate each percentage.

The basic structure is:

$$\text{Crosstab} = \text{Two categorical variables} + \text{Counts and/or percentages}$$

For the movie example:

$$\text{Opening Performance} \longleftrightarrow \text{Overall Movie Performance}$$

The crosstab therefore provides a simple way to investigate whether the two categorical variables are statistically associated.

19 Fundamental Types of Probability

The lectures discuss three fundamental types of probability:

1. Marginal (Unconditional) Probability
2. Joint Probability
3. Conditional Probability

These concepts are commonly illustrated using a **contingency table**, which organizes observations according to two or more categorical variables.

In this example, the performance of the **BITSOM Equity Fund** is observed over a 12-month period. The year is divided into two halves:

- $H1$: April–September
- $H2$: October–March

The fund's monthly returns are classified as either **positive** or **negative**.

19.1 Contingency Table

The information can be represented as follows:

	Positive Return	Negative Return	Total
<i>H1</i>	4	2	6
<i>H2</i>	5	1	6
Total	9	3	12

This table is the foundation for understanding marginal, joint, and conditional probabilities.

19.2 1. Marginal (Unconditional) Probability

Definition:

Marginal probability is the probability of a single event occurring without considering any other variable or condition.

In simple words, it answers:

“How likely is this event overall?”

The key idea is that we consider the **entire population of observations**. Therefore, the denominator is always the **grand total**.

Example: Probability of a Positive Return

There are 9 positive-return months out of 12 total months. Therefore,

$$P(\text{Positive}) = \frac{\text{Total Positive Months}}{\text{Grand Total Months}} = \frac{9}{12} = 0.75$$

Thus,

$$P(\text{Positive}) = 0.75 = 75\%$$

Conceptual intuition:

Imagine putting all 12 months into a box and randomly selecting one month. We are not told whether the month belongs to *H1* or *H2*. We simply ask:

“What is the probability that the selected month is positive?”

Since 9 of the 12 months are positive, the answer is 9/12.

Important: Marginal probability ignores the other variable.

19.3 2. Joint Probability

Definition:

Joint probability measures the probability that **two events occur together**.

The operative word is:

AND

Mathematically, the joint probability of events *A* and *B* is written as:

$$P(A \cap B)$$

Here, $\cap$ means “intersection” or “AND.”

The denominator is still the **grand total**, because we are asking about the probability of the joint event among all observations.

Example: Positive Return AND $H2$

From the contingency table, there are 5 months that satisfy both conditions:

- The month is in $H2$, and
- The fund has a positive return.

Therefore,

$$P(\text{Positive} \cap H2) = \frac{\text{Positive Months in } H2}{\text{Grand Total Months}} = \frac{5}{12}$$

Hence,

$$P(\text{Positive} \cap H2) = \frac{5}{12} \approx 0.417 = 41.7\%$$

Conceptual intuition:

Suppose you randomly select one of the 12 months.

You want the selected month to satisfy **both** conditions:

Positive Return **AND** $H2$

Only 5 months satisfy both conditions. Therefore, the probability is:

$$\frac{5}{12}$$

Notice that we have **not** changed the denominator. We are still considering all 12 possible months.

19.4 3. Conditional Probability

Definition:

Conditional probability measures the probability of an event occurring **given that another event is already known to have occurred**.

It is written as:

$$P(A \mid B)$$

and read as:

“Probability of A given B”

The most important conceptual idea is:

The information about B changes the sample space.

The general formula is:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

When working directly with a contingency table, this becomes:

$$P(A | B) = \frac{\text{Number satisfying both } A \text{ and } B}{\text{Total number satisfying } B}$$

Example: Positive Return Given $H2$

Suppose we already know that the selected month belongs to $H2$.

We are no longer considering all 12 months.

Our sample space has now shrunk to:

$$H2 = \{6 \text{ months}\}$$

Among these 6 months, 5 have positive returns.

Therefore,

$$P(\text{Positive} | H2) = \frac{\text{Positive Months in } H2}{\text{Total Months in } H2} = \frac{5}{6}$$

Thus,

$$P(\text{Positive} | H2) = \frac{5}{6} \approx 0.833 = 83.3\%$$

Why is this different from the joint probability?

This is one of the most important distinctions to understand.

For joint probability:

$$P(\text{Positive} \cap H2) = \frac{5}{12} = 41.7\%$$

We ask:

“What is the probability that a randomly selected month is BOTH positive AND in $H2$?”

Therefore, all 12 months remain possible.

For conditional probability:

$$P(\text{Positive} | H2) = \frac{5}{6} = 83.3\%$$

We ask:

“If I already know that the month is in $H2$, what is the probability that it is positive?”

Now only the 6 months in $H2$ are possible.

Therefore, the denominator changes from:

$$12 \longrightarrow 6$$

This is the fundamental idea behind conditional probability.

19.5 The “Shrinking Denominator” Concept

A useful way to remember conditional probability is:

Conditional information shrinks the sample space.

For example:

Probability	Sample Space	Denominator
$P(\text{Positive})$	All months	12
$P(\text{Positive} \cap H2)$	All months	12
$P(\text{Positive} \mid H2)$	Only $H2$ months	6

This is why a common mistake is to calculate:

$$P(\text{Positive} \mid H2) = \frac{5}{12}$$

This is incorrect because the condition $H2$ tells us that only 6 months are relevant. The correct calculation is:

$$P(\text{Positive} \mid H2) = \frac{5}{6}$$

19.6 4. Union of Events

The union of two events represents the situation in which:

A occurs OR B occurs OR both occur

It is written as:

$$P(A \cup B)$$

The standard formula is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The reason for subtracting the joint probability is to avoid **double-counting** the overlap.

Example

Suppose:

- 4 months had returns greater than 3%,
- 6 months belonged to $H2$,

- 2 months satisfied both conditions.

Let:

$$A = \{\text{return} > 3\%\}$$

and

$$B = \{H2\}$$

Then:

$$P(A) = \frac{4}{12}$$

$$P(B) = \frac{6}{12}$$

and

$$P(A \cap B) = \frac{2}{12}$$

Therefore,

$$\begin{aligned} P(A \cup B) &= \frac{4}{12} + \frac{6}{12} - \frac{2}{12} \\ &= \frac{8}{12} = \frac{2}{3} \approx 66.7\% \end{aligned}$$

$P(A \cup B) \approx 66.7\%$

Conceptual intuition:

If we simply calculated

$$\frac{4}{12} + \frac{6}{12} = \frac{10}{12},$$

the 2 months that belong to both groups would be counted twice.

Therefore, we subtract the overlap once:

$$4 + 6 - 2 = 8.$$

This gives the correct number of unique months.

19.7 5. Complement of an Event

The complement of an event A represents everything that is **not** in A .

It is written as:

$$A^c$$

The probability of an event and its complement must add up to 1:

$$P(A) + P(A^c) = 1$$

Therefore,

$$P(A^c) = 1 - P(A)$$

Example

If:

$$P(\text{Positive}) = 0.75,$$

then:

$$P(\text{Positive}^c) = 1 - 0.75 = 0.25$$

Hence,

$$P(\text{Negative or Zero Return}) = 25\%$$

Conceptual intuition:

An outcome either belongs to event A or it does not.

Therefore:

$$\text{Event} + \text{Not Event} = 100\%$$

or mathematically:

$$P(A) + P(A^c) = 1.$$

20 Key Differences Between Marginal, Joint, and Conditional Probability

Type	Question	Denominator	Key Word
Marginal	How likely is A overall?	Grand Total	Overall
Joint	How likely are A and B together?	Grand Total	AND
Conditional	How likely is A given B?	Total for B	GIVEN

For the BITSOM Equity Fund example:

$$P(\text{Positive}) = \frac{9}{12} = 75\%$$

$$P(\text{Positive} \cap H2) = \frac{5}{12} = 41.7\%$$

$$P(\text{Positive} | H2) = \frac{5}{6} = 83.3\%$$

21 A Simple Way to Remember the Three Concepts

A useful exam technique is to look for the words in the question.

Marginal Probability

Look for words such as:

overall, total, randomly selected, irrespective of

Use the grand total as the denominator.

$$P(A) = \frac{\text{Number of } A}{\text{Grand Total}}$$

Joint Probability

Look for:

AND

Find the cell where the two events intersect.

$$P(A \cap B) = \frac{\text{Number in } A \text{ and } B}{\text{Grand Total}}$$

Conditional Probability

Look for:

given, if we know that, among

The condition determines the denominator.

$$P(A | B) = \frac{\text{Number in } A \text{ and } B}{\text{Total number in } B}$$

22 Financial Interpretation

These probability concepts are particularly useful in financial analysis because they allow us to distinguish between an overall probability and a probability under a specific market or time condition.

For example:

$$P(\text{Positive}) = 75\%$$

means that the fund had a positive return in 75% of all observed months.

However,

$$P(\text{Positive} | H2) = 83.3\%$$

means that once we restrict our attention to $H2$, the probability of a positive return increases to 83.3%.

This does **not** mean that $H2$ necessarily causes positive returns. It only describes the observed relationship in this particular sample.

This distinction is important:

Conditional probability describes probability under a condition; it does not automatically imply causation.

23 Important Exam Insight

The most common mistake when solving conditional probability questions is using the **grand total** as the denominator.

Always ask yourself:

“What information has already been given to me?”

If the question says:

$$P(A \mid B),$$

then B defines the new sample space.

Therefore:

Denominator = Total number of observations satisfying B

For example:

$$P(\text{Positive} \mid H2) = \frac{5}{6},$$

not

$$\frac{5}{12}.$$

The number 12 describes the entire year, whereas the condition $H2$ restricts us to only 6 months.

24 Quick Revision Summary

Marginal:	$P(A) = \frac{A}{\text{Total}}$
Joint:	$P(A \cap B) = \frac{A \text{ and } B}{\text{Total}}$
Conditional:	$P(A \mid B) = \frac{A \text{ and } B}{B}$

The easiest way to remember the difference is:

Marginal	→ OVERALL
Joint	→ AND
Conditional	→ GIVEN

The central idea is that **joint probability keeps the full sample space, while conditional probability changes the sample space based on the information given.**

“latex

25 Probability and Statistical Concepts

25.1 1. Marginal, Joint, and Conditional Probability

A contingency table helps distinguish between marginal, joint, and conditional probability.

For the BITSOM Equity Fund, suppose monthly returns over 12 months are classified according to the half of the year and whether the return was positive or negative.

	Positive	Negative	Total
<i>H1</i>	4	2	6
<i>H2</i>	5	1	6
Total	9	3	12

25.1.1 1.1 Marginal Probability

Marginal probability measures the probability of a single event occurring without considering another variable or condition.

It answers:

How likely is this event overall?

The denominator is the grand total:

$$P(A) = \frac{\text{Number of observations satisfying } A}{\text{Grand Total}}$$

For example, there are 9 positive-return months out of 12 months:

$$P(\text{Positive}) = \frac{9}{12} = 0.75 = 75\%.$$

Therefore,

$$P(\text{Positive}) = 75\%$$

Concept:

Marginal probability considers the entire dataset. It does not condition on whether the month belongs to *H1* or *H2*.

25.1.2 1.2 Joint Probability

Joint probability measures the probability that two events occur simultaneously.

The key word is:

AND

It is written as:

$$P(A \cap B)$$

The denominator is also the grand total:

$$P(A \cap B) = \frac{\text{Number of observations satisfying both } A \text{ and } B}{\text{Grand Total}}$$

There are 5 months that are both in $H2$ and positive-return months. Therefore,

$$P(\text{Positive} \cap H2) = \frac{5}{12} \approx 0.417 = 41.7\%.$$

Hence,

$$P(\text{Positive} \cap H2) = 41.7\%$$

Important:

Marginal and joint probabilities use the **same denominator**. They differ in their numerators.

$$P(\text{Positive}) = \frac{9}{12}$$

whereas

$$P(\text{Positive} \cap H2) = \frac{5}{12}.$$

Thus,

$$\begin{array}{l} \text{Marginal} \rightarrow \text{Row or column total} \\ \text{Joint} \rightarrow \text{Intersection cell} \end{array}$$

25.1.3 1.3 Conditional Probability

Conditional probability measures the probability of an event given that another event is already known to have occurred.

It is written as:

$$P(A \mid B)$$

and read as:

$$\text{Probability of } A \text{ given } B$$

The general formula is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

In a contingency table:

$$P(A \mid B) = \frac{\text{Number satisfying both } A \text{ and } B}{\text{Total number satisfying } B}$$

Suppose we know that the month belongs to $H2$.

There are only 6 possible months, and 5 of those months have positive returns.

Therefore,

$$P(\text{Positive} \mid H_2) = \frac{5}{6} \approx 0.8333 = 83.3\%.$$

Hence,

$$P(\text{Positive} \mid H_2) = 83.3\%$$

25.1.4 1.4 The Shrinking Denominator

The key idea behind conditional probability is that the condition changes the sample space.

Initially:

12 possible months

After learning that the month belongs to H_2 :

6 possible months

Therefore:

$$12 \longrightarrow 6.$$

This gives the important rule:

Conditional information shrinks the sample space.

25.1.5 1.5 Comparison

Type	Question	Numerator	Denominator
Marginal	How likely is A overall?	Total A	Grand Total
Joint	How likely are A and B together?	$A \cap B$	Grand Total
Conditional	How likely is A given B?	$A \cap B$	Total B

A simple memory rule is:

Marginal $\rightarrow$ OVERALL
 Joint $\rightarrow$ AND
 Conditional $\rightarrow$ GIVEN

25.2 2. Union of Events

The union of two events means that event A , event B , or both events occur.

It is represented by:

$$P(A \cup B)$$

The formula is:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The intersection is subtracted because the overlapping observations would otherwise be counted twice.

For example, suppose:

$$P(A) = \frac{4}{12}, \quad P(B) = \frac{6}{12}, \quad P(A \cap B) = \frac{2}{12}.$$

Then:

$$\begin{aligned} P(A \cup B) &= \frac{4}{12} + \frac{6}{12} - \frac{2}{12} \\ &= \frac{8}{12} = \frac{2}{3} \approx 66.7\%. \end{aligned}$$

Therefore,

$$\boxed{P(A \cup B) \approx 66.7\%}$$

25.3 3. Complement of an Event

The complement of event A represents all outcomes that are not in A .

It is written as:

$$A^c.$$

Since an event and its complement together contain all possible outcomes:

$$P(A) + P(A^c) = 1.$$

Therefore:

$$\boxed{P(A^c) = 1 - P(A)}$$

If:

$$P(\text{Positive}) = 0.75,$$

then:

$$P(\text{Positive}^c) = 1 - 0.75 = 0.25.$$

Thus,

$$\boxed{P(\text{Negative or Zero}) = 25\%}$$

25.4 4. Coefficient of Variation

25.4.1 4.1 Definition

The **Coefficient of Variation (CV)** is a relative measure of dispersion.

It compares the standard deviation of a dataset with its mean.

For a sample:

$$\boxed{CV = \frac{s}{\bar{x}} \times 100\%}$$

For a population:

$$CV = \frac{\sigma}{\mu} \times 100\%$$

where:

s = Sample standard deviation,

$\bar{x}$ = Sample mean,

σ = Population standard deviation,

μ = Population mean.

25.4.2 4.2 Conceptual Meaning

Standard deviation answers:

How much do the observations fluctuate?

CV answers:

How much volatility am I accepting per unit of average return?

Therefore:

$$CV = \frac{\text{Risk}}{\text{Average Return}}$$

A lower CV generally indicates greater consistency relative to the average return, assuming the mean is positive and meaningful for comparison.

A higher CV indicates greater volatility relative to the average return.

25.4.3 4.3 BITSOM Equity Fund

Suppose:

$$\bar{x}_E = 1.88\%$$

and

$$s_E = 2.33\%.$$

Then:

$$CV_E = \frac{2.33\%}{1.88\%} \times 100\% \approx 123.9\%.$$

Therefore:

$$CV_E \approx 123.9\%$$

25.4.4 4.4 BITSOM Debt Fund

Suppose:

$$\bar{x}_D = 0.58\%$$

and

$$s_D = 0.21\%.$$

Then:

$$CV_D = \frac{0.21\%}{0.58\%} \times 100\% \approx 36.2\%.$$

Therefore:

$$\boxed{CV_D \approx 36.2\%}$$

25.4.5 4.5 Interpretation

The Equity Fund has:

$$CV_E = 123.9\%$$

while the Debt Fund has:

$$CV_D = 36.2\%.$$

Although the Equity Fund has the higher average monthly return:

$$1.88\% > 0.58\%,$$

its volatility is much larger relative to its average return.

Thus, based on relative risk per unit of average return:

The Debt Fund is more consistent than the Equity Fund.

25.5 5. Practice Question: Coefficient of Variation

Consider two funds:

	Mean Return	Standard Deviation
Fund Alpha	3.50%	1.05%
Fund Beta	3.50%	4.20%

For Fund Alpha:

$$CV_{\text{Alpha}} = \frac{1.05\%}{3.50\%} \times 100\% = 30\%.$$

Therefore:

$$\boxed{CV_{\text{Alpha}} = 30\%}$$

For Fund Beta:

$$CV_{\text{Beta}} = \frac{4.20\%}{3.50\%} \times 100\% = 120\%.$$

Therefore:

$$CV_{\text{Beta}} = 120\%$$

Since both funds have the same mean return:

$$3.50\% = 3.50\%,$$

the CV clearly reveals the difference in relative risk.

Since:

$$30\% < 120\%,$$

we conclude:

Fund Alpha is more efficient on a relative-risk basis.

25.6 6. Calinski–Harabasz Index

25.6.1 6.1 Definition

The **Calinski–Harabasz (CH) Index** is a clustering evaluation metric used to assess the quality of a clustering solution.

It is commonly used to help determine the appropriate number of clusters k in algorithms such as K-means.

The formula is:

$$CH(k) = \frac{BCSS/(k-1)}{WCSS/(n-k)}$$

where:

$BCSS$ = Between-Cluster Sum of Squares,

$WCSS$ = Within-Cluster Sum of Squares,

n = Total number of observations,

k = Number of clusters.

25.6.2 6.2 Between-Cluster Sum of Squares

BCSS measures how separated the clusters are from one another.

A large BCSS indicates:

Clusters are well separated.

Therefore, we generally prefer:

$$BCSS \uparrow .$$

25.6.3 6.3 Within-Cluster Sum of Squares

WCSS measures the total squared distance between observations and their assigned cluster centroids.

A small WCSS indicates:

Observations within each cluster are close to one another.

Therefore, we generally prefer:

$$WCSS \downarrow .$$

Thus, a good clustering solution should have:

BCSS high and *WCSS* low.

25.6.4 6.4 Practice Calculation

Suppose:

$$n = 8, \quad k = 3,$$

with:

$$BCSS = 4571.75$$

and:

$$WCSS = 304.00.$$

First:

$$\frac{BCSS}{k-1} = \frac{4571.75}{3-1} = 2285.875.$$

Next:

$$\frac{WCSS}{n-k} = \frac{304}{8-3} = 60.8.$$

Therefore:

$$CH(3) = \frac{2285.875}{60.8} \approx 37.60.$$

Hence:

$$CH(3) \approx 37.60$$

Suppose:

k	CH Index
2	23.24
3	37.60
4	33.28

The highest CH Index occurs at:

$$k = 3.$$

Therefore:

$$k = 3 \text{ is the preferred clustering solution.}$$

The key exam rule is:

$$\text{Higher CH Index} = \text{Better clustering solution}$$

25.7 7. Z-Score

25.7.1 7.1 Definition

A Z-score measures how many standard deviations an observation is above or below the mean.

For a sample:

$$Z = \frac{x - \bar{x}}{s}$$

For a population:

$$Z = \frac{x - \mu}{\sigma}$$

where:

x = Observed value,
 $\bar{x}$ = Sample mean,
 s = Sample standard deviation,
 μ = Population mean,
 σ = Population standard deviation.

25.7.2 7.2 Conceptual Meaning

The quantity:

$$x - \bar{x}$$

measures the raw distance between an observation and the mean.

Dividing by the standard deviation converts this distance into standardized units.

Therefore:

$$Z = \text{Distance from the mean measured in standard deviations}$$

For example:

$$Z = 2$$

means the observation is 2 standard deviations above the mean.

Similarly:

$$Z = -3$$

means the observation is 3 standard deviations below the mean.

25.7.3 7.3 Practice Example

Suppose:

$$\bar{x} = 1.31\%$$

and:

$$s = 2.71\%.$$

During March 2020, the observed return was:

$$x = -11.20\%.$$

Then:

$$\begin{aligned} Z &= \frac{-11.20\% - 1.31\%}{2.71\%} \\ &= \frac{-12.51\%}{2.71\%} \\ &\approx -4.62. \end{aligned}$$

Therefore:

$$\boxed{Z \approx -4.62}$$

The negative sign indicates that the observation is below the mean.

The magnitude:

$$|Z| = 4.62$$

indicates that the observation is approximately 4.62 standard deviations below the mean.

25.7.4 7.4 Outlier Detection

A common rule of thumb is:

$$\boxed{|Z| > 3 \Rightarrow \text{Potential Outlier}}$$

Since:

$$|-4.62| > 3,$$

the observation should be flagged for investigation.

However:

$$\boxed{\text{Potential Outlier} \neq \text{Automatically Delete}}$$

A large financial return may be caused by:

a data-entry error,
a measurement problem,
or a genuine market shock.

If the observation represents a genuine market event, deleting it could remove valuable information about extreme market risk.

25.8 8. p-Value

25.8.1 8.1 Definition

A p-value measures how compatible an observed test statistic is with the null hypothesis.

More precisely, it is the probability of obtaining a test statistic at least as extreme as the observed statistic, assuming that the null hypothesis is true.

The key principle is:

$$\boxed{\text{Small p-value} \Rightarrow \text{Evidence against } H_0}$$

Using the standard significance level:

$$\alpha = 0.05,$$

the decision rule is:

$$\boxed{p < 0.05 \Rightarrow \text{Reject } H_0}$$

and:

$$\boxed{p \geq 0.05 \Rightarrow \text{Fail to reject } H_0}$$

Important:

We generally say “fail to reject” rather than “accept the null hypothesis.”

25.8.2 8.2 Individual Regression Coefficient

For an individual regression coefficient β_j , the null hypothesis is typically:

$$H_0 : \beta_j = 0.$$

The alternative hypothesis may be:

$$H_1 : \beta_j \neq 0.$$

If:

$$p_j < 0.05,$$

we reject:

$$H_0 : \beta_j = 0.$$

Therefore, there is statistical evidence that the predictor is associated with the dependent variable, conditional on the model specification.

25.8.3 8.3 Overall F-Test

The overall F-test examines whether the regression model has explanatory power collectively.

A common null hypothesis is:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0.$$

The alternative hypothesis is:

$$H_1 : \text{At least one } \beta_j \neq 0.$$

If:

$$p_F < 0.05,$$

we reject the null hypothesis and conclude that the model is statistically significant overall.

25.8.4 8.4 Practice Example

Suppose a regression model predicts the monthly return of the BITSOM Equity Fund using:

$$X_1 = \text{Nifty 50 Return},$$

$$X_2 = \text{G-Sec Yield Change},$$

and:

$$X_3 = \text{USD/INR Change}.$$

Suppose the regression output gives:

$$p_F < 0.001,$$

$$p_{X_1} < 0.001,$$

$$p_{X_2} < 0.001,$$

and:

$$p_{X_3} = 0.120.$$

Using:

$$\alpha = 0.05,$$

we interpret the results as follows.

Overall Model

Since:

$$p_F < 0.001 < 0.05,$$

we reject the overall null hypothesis.

Therefore:

The regression model is statistically significant overall.

Nifty 50

Since:

$$p_{X_1} < 0.001 < 0.05,$$

we reject:

$$H_0 : \beta_1 = 0.$$

Therefore:

X_1 is statistically significant at the 5% level.

G-Sec Change

Since:

$$p_{X_2} < 0.001 < 0.05,$$

we reject:

$$H_0 : \beta_2 = 0.$$

Therefore:

X_2 is statistically significant at the 5% level.

USD/INR Change

Since:

$$p_{X_3} = 0.120 > 0.05,$$

we fail to reject:

$$H_0 : \beta_3 = 0.$$

Therefore:

X_3 is not statistically significant at the 5% level.

25.8.5 8.5 Important Interpretation of p-Value

A p-value does **not** represent:

$$P(H_0 \text{ is true}).$$

For example, if:

$$p = 0.03,$$

we should not say:

There is a 3% probability that H_0 is true.

Instead, we say that the observed result would be relatively unusual under the assumption that H_0 is true, providing evidence against H_0 .

25.9 9. Final Exam Cheat Sheet

Marginal Probability:	$P(A) = \frac{\text{Number of } A}{\text{Grand Total}}$
Joint Probability:	$P(A \cap B) = \frac{\text{Number of } A \text{ and } B}{\text{Grand Total}}$
Conditional Probability:	$P(A B) = \frac{\text{Number of } A \text{ and } B}{\text{Number of } B}$
Union:	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Complement:	$P(A^c) = 1 - P(A)$
Coefficient of Variation:	$CV = \frac{s}{\bar{x}} \times 100\%$
CH Index:	$CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}$
Z-score:	$Z = \frac{x - \bar{x}}{s}$
p-value decision:	$p < \alpha \Rightarrow \text{Reject } H_0$

25.10 10. Key Exam Rules

1. Marginal and joint probabilities use the same denominator: the grand total.
2. Marginal probability uses a row or column total as its numerator.
3. Joint probability uses the intersection cell as its numerator.
4. Conditional probability changes the denominator because the condition changes the sample space.

5. Remember:

AND $\rightarrow$ Joint

GIVEN $\rightarrow$ Conditional

OR $\rightarrow$ Union

NOT $\rightarrow$ Complement

6. **CV** measures relative risk per unit of average return.
7. **CH Index: higher is better.**
8. **Z-score** measures the distance from the mean in standard deviation units.
9. $|Z| > 3$ is a common rule for flagging a potential outlier.
10. **A potential outlier should not automatically be deleted.**
11. **p-value** < 0.05 provides evidence against the null hypothesis at the 5% significance level.
12. **Statistical significance does not automatically imply economic or practical significance.**

26 Confusion Matrix Classification Evaluation Metrics

26.1 What is a Confusion Matrix?

Every classification metric—whether for logistic regression, decision trees, or support vector machines—is built on top of the **confusion matrix**. A confusion matrix is a structured table that compares the **actual (true) outcomes** in a dataset against the **predicted outcomes** generated by a machine learning model.

The matrix is structured along two axes:

- **Rows** represent the **actual class** of each observation.
- **Columns** represent the **predicted class** determined by the algorithm.

26.2 The Four Core Cells

In financial classification tasks, such as credit risk modeling, the conventional labeling is:

- **Positive (1):** The borrower defaulted on a bank loan.
- **Negative (0):** The borrower paid on time and did not default.

Note: “Positive” is simply an analytical label for the class of interest; it does not represent a positive real-world event.

Consider a bank dataset containing 10,000 historical loan customers, of whom 100 defaulted and 9,900 did not. The resulting confusion matrix is:

Actual / Predicted		Predicted
True Positive (TP): 95	Actually defaulted; predicted to default	False Negative (FN): 5 A
	True Negative (TN): 9,700 Paid on time; predicted to pay	9,900

Table 4: Confusion Matrix for Loan Default Classification

The four cells can therefore be interpreted as follows:

- **True Positive (TP):** The borrower actually defaulted and the model correctly predicted default.
- **False Negative (FN):** The borrower actually defaulted, but the model incorrectly predicted that the borrower would pay.
- **False Positive (FP):** The borrower paid on time, but the model incorrectly predicted default.
- **True Negative (TN):** The borrower paid on time and the model correctly predicted that the borrower would pay.

26.3 Accuracy

Formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Observations}}$$

Example:

Using the confusion matrix above:

$$\text{Accuracy} = \frac{95 + 9,700}{10,000}$$

$$\text{Accuracy} = 97.95\%$$

The Accuracy Trap:

Accuracy can be highly misleading when evaluating **imbalanced datasets**, where one class is much more common than the other. In this example, only 1 of borrowers defaulted.

A trivial model that predicts “no default” for every applicant would still achieve:

$$\text{Accuracy} = \frac{9,900}{10,000}$$

$$\text{Accuracy} = 99\%$$

However, this model would identify **zero defaults**. Therefore, accuracy should always be evaluated alongside precision, recall, and other appropriate classification metrics.

26.4 Recall (Sensitivity)

Recall measures the proportion of all actual positive cases that the model successfully identifies.

Formula:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Example:

$$\text{Recall} = \frac{95}{95 + 5}$$

$$\text{Recall} = 95.0\%$$

Recall focuses on the **actual positive class**. It is particularly important when failing to identify a positive case carries a high cost.

26.5 Precision

Precision measures the proportion of predicted positive cases that are actually positive.

Formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Example:

$$\text{Precision} = \frac{95}{95 + 200}$$

$$\text{Precision} = 32.2\%$$

Precision focuses on the **predicted positive class**. It becomes particularly important when false alarms or unnecessary interventions carry a significant cost.

26.6 F1 Score

The F1 score is the **harmonic mean of precision and recall**. Unlike a simple arithmetic average, the harmonic mean penalizes cases where one of the two metrics is substantially lower than the other.

Formula:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Example:

Using

$$\text{Precision} = 0.322$$

and

$$\text{Recall} = 0.950$$

we obtain:

$$\text{F1 Score} = 2 \times \frac{0.322 \times 0.950}{0.322 + 0.950}$$

$$\text{F1 Score} = 48.1\%$$

The relatively low F1 score, despite the high recall, highlights the trade-off created by the large number of false positives.

27 Precision–Recall Trade-off in Finance

The appropriate classification threshold depends fundamentally on the relative financial costs of a **False Positive** and a **False Negative**.

Because of the **precision–recall trade-off**, improving one metric often comes at the expense of the other.

High Recall Lower Classification Threshold

Catches more defaults: Low FN but increases false alarms: High FP
Higher Classification Threshold High Precision
Reduces false alarms: Low FP but may miss defaults: High FN

27.1 Case A: The Risk-Averse Lender – Prioritizing Recall

Business Focus: Minimizing direct credit losses.

Cost Analysis: Extending credit to a borrower who subsequently defaults represents a **False Negative** and can result in substantial financial losses, including loss of the loan principal.

By contrast, rejecting a creditworthy borrower represents a **False Positive** and primarily results in a lost opportunity to earn interest or other fees.

Strategy: The lender lowers the classification threshold. The objective is to identify as many potential defaulters as possible, even if this means flagging some creditworthy applicants.

This strategy generally produces:

- Higher True Positives,
- Fewer False Negatives,
- Higher Recall, and
- Potentially lower Precision because of increased False Positives.

Real-world Example: MSME loan-underwriting platforms or credit-card issuers may adopt a more conservative risk threshold during economic downturns when preventing charge-offs becomes a priority.

27.2 Case B: The Growth-Focused Acquisition Desk – Prioritizing Precision

Business Focus: Minimizing lost business opportunities and customer friction.

Cost Analysis: Suppose a model incorrectly predicts that a reliable, high-net-worth customer will default. This **False Positive** may cause the bank to deny a loan, potentially damaging the customer relationship, encouraging the customer to move to a competitor, and eliminating future fee and interest income.

Strategy: The bank raises the classification threshold and flags an applicant as high risk only when the evidence of default is sufficiently strong.

This strategy generally produces:

- Fewer False Positives,
- Higher Precision, and
- Potentially lower Recall because of increased False Negatives.

Real-world Example: Premium or ultra-high-net-worth wealth-management segments and personal-loan cross-selling systems may place greater emphasis on avoiding unnecessary customer friction.

27.3 Case C: Real-Time Fraud Prevention – A Balanced Objective

Business Focus: Maintaining a practical balance between customer convenience and transaction security.

Fraud detection illustrates the opposing costs particularly well:

- **False Negative (Missed Fraud):** A fraudulent transaction worth 10,00,000 may be approved, resulting in substantial financial losses or write-offs.
- **False Positive (False Alarm):** A legitimate customer's card may be blocked during a 500 purchase, creating significant inconvenience and potentially damaging customer trust.

Strategy: The product team may use the **F1 score** as one of the evaluation criteria for selecting a classification threshold. The objective is to maintain strong fraud detection performance while avoiding an excessive number of customer-facing false alarms.

Real-world Example: Real-time transaction-monitoring systems used by payment gateways and financial technology platforms such as Paytm, PhonePe, and Razorpay must continuously balance fraud prevention against customer experience.

28 Key Takeaway

The confusion matrix provides the foundation for understanding classification performance. Its four fundamental outcomes—

True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN)—allow us to derive metrics such as accuracy, precision, recall, and the F1 score.

In financial applications, however, there is no universally optimal metric. The appropriate evaluation criterion depends on the underlying business objective and, more importantly, on the relative costs of different types of classification errors.

Therefore, selecting a classification threshold should be treated as a **business decision supported by statistical evidence**, rather than simply as a mathematical optimization problem.

29 Decentralized Autonomous Organizations (DAOs)

A **DAO** stands for **Decentralized Autonomous Organization**. It is an internet-native organization that operates without a centralized management team, relying instead on **smart contracts** and **community voting** to make decisions.

29.1 Core Characteristics of a DAO

DAOs are generally characterized by the following features:

- **On-Chain Rules:** The rules of the organization are written directly into smart contract code deployed on a blockchain.

- **Token-Based Governance:** Decisions are typically made by **governance token holders**, such as holders of the AAVE token in the Aave protocol, rather than by executives or a traditional board of directors.
- **Scope of Management:** DAOs may govern protocol upgrades, token emission rates, grants, risk parameters, and treasury allocations.

29.2 How DAO Decisions and Rule Changes Work

In a DAO, governance rules and protocol parameters are generally not intended to be modified unilaterally by a single participant. When the community wants to change a coded parameter—such as an insurance payout threshold or an interest rate—the proposed change typically passes through a formal governance workflow.

Step 1 – Submission:

A participant submits a formal governance proposal describing the proposed change.

Step 2 – Discussion:

The community discusses the proposal, evaluates its potential benefits and risks, and provides feedback.

Step 3 – Voting:

Eligible token holders cast their votes using the DAO's governance mechanism.

Step 4 – Execution:

If the proposal satisfies the required voting conditions, the approved change can be executed on-chain through the relevant smart contracts.

Step 5 – Monitoring:

After execution, the community monitors the resulting changes and their effects on the protocol.

The overall governance process can therefore be summarized as:

Submission → Discussion → Voting → Execution → Monitoring

Important Exam Point:

Rules in a DAO are not necessarily changed instantly by a single person simply because that person controls a 51 stake. The actual governance mechanism depends on the DAO's smart contracts and voting rules, but where formal token-holder governance is required, a proposal and voting process must be followed before an approved change can be executed.

29.3 DAO Governance vs. Traditional Governance

The governance structures of traditional organizations and DAOs differ in several important ways.

The fundamental distinction is that traditional organizations rely primarily on **hierarchical authority and institutional processes**, whereas DAOs rely more heavily on **code, token-based governance, and transparent on-chain processes**.

29.4 Key Risks and Challenges of DAOs

Although DAOs reduce dependence on centralized management, they do not eliminate the need for oversight. Instead, they introduce several distinct governance and operational challenges.

Aspect	Traditional Banking Governance	DAO Governance Who Governs
Boards, executives, regulators, and committees	Token holders, delegates, and smart contracts	Minutes, decisions, and disclosures may remain largely internal
Proposals, votes, and transactions are generally visible on-chain	Manual approvals and internal organizational systems	Smart contracts can execute approved actions automatically on-chain
Execution		

Table 5: Traditional Governance vs. DAO Governance

29.4.1 1. Whale Concentration

Large token holders, commonly referred to as **whales**, may possess a disproportionately large amount of voting power.

If voting power is closely tied to token ownership, a small number of large holders may exert substantial influence over governance decisions.

Large Token Holdings → High Voting Power → Potential Governance Concentration

This creates a potential tension between the decentralized structure of a DAO and the concentration of economic power among a small number of participants.

29.4.2 2. Voter Apathy

Not every token holder participates actively in governance.

Some participants may lack the time, expertise, or incentive to review proposals and vote on them. Low participation can therefore reduce the quality of collective decision-making and increase the relative influence of highly active participants.

Low Participation → Smaller Active Voter Base → Greater Relative Influence of Active Participants

29.4.3 3. Legal and Regulatory Uncertainty

DAOs can operate through globally distributed blockchain networks, creating difficult legal and regulatory questions.

Important issues may include:

- Which jurisdiction has authority over the DAO?
- Who is legally responsible for decisions made by the DAO?
- How should governance tokens be classified under applicable law?
- What regulatory requirements apply to decentralized financial protocols?

Consequently, DAOs continue to face significant legal and regulatory uncertainty across different jurisdictions.

30 Key Takeaway

A DAO replaces much of the traditional organizational hierarchy with a combination of **smart contracts, blockchain-based transparency, and community governance**.

The basic governance cycle is:

Proposal → Discussion → Vote → Execution

However, decentralization does not automatically eliminate governance risks. **Whale concentration, voter apathy, and legal uncertainty** remain important challenges.

Therefore, the key idea to remember is:

DAO = Code + Token Governance + Community Decision-Making